Zero-density estimates from an improved large-values bound

A source-faithful mathematical blueprint of the registered Lean development

Mathematics formalized by Mark Lewko

Informal reconstruction for study; not an author-approved paper

Source: Palomar entry PALOMAR-2026-10-11-000010, version 1

Scope, provenance, and reading conventions

This document reconstructs the mathematical argument represented by the four endpoint theorems in `lean/Solution.lean` of `mrklwk/zeta-density-39-over-17-lean`, at the exact commit

`ee2a8a32be49a4f73c7386e9dffdea24d7e3d8c0`.

The immutable record is `https://data.palomar-registry.org/entries/PALOMAR-2026-10-11-000010-v1.json`; the requested entry is `https://palomar-registry.org/entry?id=PALOMAR-2026-10-11-000010&version=1`. The dependency on mathlib is pinned to `065356127b1dc0016f66b7283ce0ce2c4055aa55`. All references to a source file mean this snapshot, not a changing default branch.

What is reconstructed. The text gives full proofs of the endpoint composition, the middle-range scale optimization and absorption, the passage from truncated moments to separated values, the two new density windows, multiplicity restoration, dyadic summation, and uniformization. It also spells out the operators, Fourier masks, affine averages, detector families, and imported analytic interfaces needed by those proofs. The substantial previously established analytic machinery is explicitly identified as such: an invocation of an established input below is not a new assumption in an endpoint theorem. In particular, a proof in this blueprint is understood relative to the precisely stated, already proved inputs that it cites. The native fourth-moment, point-mean/stationary-phase, classical reflection, and Fourier-affine foundations are not represented as if they followed from elementary algebra. Their actual role and source interfaces are recorded separately.

Boundary of the transcription. This is a mathematical blueprint, not a literal translation of every auxiliary lemma in all 824 locally imported modules. There are supporting library proofs behind the established inputs. These are not re-proved down to the foundations of measure theory, special functions, or the native stationary-source ports here. The accompanying complete pinned source archive preserves those proofs. Source correspondences make this boundary explicit, rather than concealing it with phrases such as "standard estimates." No independent verification of a fresh full Lean build is claimed by this reconstruction.

Fidelity rules. All real parameters in the endpoint statements are retained as real parameters. The interval of source integers is exactly $(N, 2N]$, implemented with natural floors. Both signs of zero ordinates and both closed height endpoints are retained. Zero locations and zero multiplicities are never identified. A fixed constant is selected before every variable over which the corresponding result is uniform. The rightmost density branch starts *strictly* above $3/4$. We do not assert a globally uniform constant for the exact discontinuous curve. The historical name `CandidateMiddleRangeEstimate` denotes a proposition that is

proved; conversely, the separately defined `ShortMobiusFourthMoment` is not silently used as a theorem.

Logical and analytic conventions. Integrals on $\mathbb{R}$ use Lebesgue measure unless another measure is displayed. The Fourier transform is

$$\widehat{f}(\xi) = \int_{\mathbb{R}} \mathrm{e}^{-2\pi \mathrm{i}\xi t} f(t)\, dt, \qquad e(x) = \mathrm{e}^{2\pi \mathrm{i} x}, \qquad \lambda(x) = \frac{\log x}{2\pi} \quad (x > 0).$$

Its Plancherel constant is one. Complex inner products are conjugate-linear in the first argument. A real power of a positive real base means $x^a = \exp(a \log x)$. Every application below of monotonicity in an exponent has base at least one. "Eventually in H" means that one height threshold exists before all subsequently quantified variables. A notation $\ll_\varepsilon$ abbreviates an existential positive constant independent of all other displayed varying objects, not an effective numerical value.

Proof automation. Source uses of `ring`, `field_simp`, `norm_num`, `linarith`, `nlinarith`, and `gcongr` are expanded as algebra, positive-denominator comparisons, or monotonicity. They are not additional analytic inputs. Classical finite-set selection and Hilbert-basis selection account for uses of choice. The registered axiom allowance is `propext`, `Classical.choice`, and `Quot.sound`. The holes in the independent `Challenge.lean` specification are not imported by `Solution.lean` and are not premises of the proof.

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Chapter 1

The objects counted and the four endpoint statements

1.1 Actual zeros and analytic multiplicity

Definition 1.1. Let ζ be mathlib's Riemann zeta function. A nontrivial zero is a complex number ρ for which

$$\zeta(\rho) = 0, \qquad 0 < \operatorname{Re}\rho < 1.$$

Define $m(\rho)$ to be the natural analytic vanishing order `analyticOrderNatAtriemannZeta` at ρ. For real σ, a, b, set

$$Z(\sigma; a, b) = \{\rho : \zeta(\rho) = 0,\ 0 < \operatorname{Re}\rho < 1,\ \sigma \le \operatorname{Re}\rho,\ a \le \operatorname{Im}\rho \le b\}.$$

Define the slab and density counts

$$n(\sigma; a, b) = \sum_{\rho \in Z(\sigma;a,b)} m(\rho), \qquad N(\sigma, T) = n(\sigma; -T, T).$$

Lean correspondence: `MathCollab.Density.ZetaCounting:IsNontrivialZetaZero,zetaMultiplicity,zetaZeroRegion,zetaZeroFinset,zetaSlabCount,zetaDensityCount`.

Lemma 1.2 (The finite-set and multiplicity conventions are genuine)**.** *Every set $Z(\sigma; a, b)$ is finite. At a nontrivial zero, $m(\rho)$ is finite and positive, and there is an analytic function g near ρ, with $g(\rho) \neq 0$, such that locally*

$$\zeta(z) = (z - \rho)^{m(\rho)} g(z).$$

Proof. For $\rho \in Z(\sigma; a, b)$, the source bounds $|\operatorname{Im}\rho| \le |a| + |b|$ and

$$|\rho| \le |\operatorname{Re}\rho| + |\operatorname{Im}\rho| \le 1 + |a| + |b|.$$

Thus the zero set is contained in the intersection of a closed ball and the actual zeta-zero set. The imported mathlib theorem that this intersection is finite gives the required finite set, without assuming that the open strip is compact. Zeta is analytic away from 1. If its analytic order at any $\rho \neq 1$ were infinite, it would vanish on a neighborhood of ρ. The identity theorem on the connected punctured plane $\mathbb{C} \setminus \{1\}$ would make it identically zero there. This contradicts its nonvanishing at 2. The finite-order local factorization theorem for analytic functions then gives g. At a zero the order is not zero, so its conversion to a natural number is strictly positive. □

Lean correspondence: `MathCollab.Density.ZetaCounting:zetaZeroRegion_finite,zeta_analyticOrder_ne_top,zetaMultiplicity_pos,zetaMultiplicity_local_factorization`.

Lemma 1.3 (Monotonicity, inclusions, and the right edge)**.** *If $\sigma' \le \sigma$, $a' \le a$, and $b \le b'$, then $n(\sigma; a, b) \le n(\sigma'; a', b')$. Consequently if also $T \le T'$, then $N(\sigma, T) \le N(\sigma', T')$. If $\sigma \ge 1$, $N(\sigma, T) = 0$. The number of zero locations in a slab is at most its multiplicity count.*

Proof. The inequalities on parameters give an inclusion of zero sets. Sum the nonnegative natural multiplicities over that inclusion. If $\sigma \ge 1$, a putative zero has both $\operatorname{Re}\rho \ge \sigma \ge 1$ and $\operatorname{Re}\rho < 1$, a contradiction. For the last assertion sum $1 \le m(\rho)$ over the finite zero set. □

Lean correspondence: `MathCollab.Density.ZetaCounting:zetaSlabCount_mono,zetaDensityCount_mono,zetaZeroFinset_card_le_count;GuthMaynard.Improvement.GlobalDensityConditional:zetaDensityCount_zero_of_one_le`.

1.2 Source polynomials and separated observation sets

Definition 1.4. For $N \in \mathbb{R}$ put

$$I_N = \{n \in \mathbb{N} : \lfloor N \rfloor_+ < n \le \lfloor 2N \rfloor_+\}, \qquad D_{N,a}(t) = \sum_{n \in I_N} a_n \mathrm{e}^{\mathrm{i}t\log(n/N)}.$$

For $N \ge 0$, membership in I_N is equivalent to $N < n \le 2N$. A finite set $W \subset \mathbb{R}$ is one-separated if $|t - u| \ge 1$ whenever $t, u \in W$ and $t \ne u$.

The coefficient hypothesis throughout is $|a_n| \le 1$ for $n \in I_N$; no hypothesis is placed on coefficients outside that interval.

Lean correspondence: `GuthMaynard.Improvement.Target:boundedPolynomial;SourceRatioAssembly:sourceRatioIndices_mem`.

Lemma 1.5 (Normalization and real-to-integer source transfer)**.** *For $N > 0$, the norm of $D_{N,a}$ equals the norm of $\sum_{n \in I_N} a_n \mathrm{e}^{\mathrm{i}t\log n}$. For $N \ge 2$ and $M = \lfloor N \rfloor_+$,*

$$M \ge 2, \quad N/2 \le M \le N, \quad \lfloor 2N \rfloor_+ \in \{2M, 2M+1\}.$$

For bounded source coefficients,

$$|D_{N,a}(t)| \le |D_{M,a}(t)| + 1.$$

In particular a value at least $V \ge 2$ transfers to the integer-length polynomial at threshold $V/2$.

Proof. Factor out $\mathrm{e}^{-\mathrm{i}t\log N}$, which has modulus one. The floor inequalities $M \le N < M+1$ give the bounds on M and the two possible values of $\lfloor 2N \rfloor_+$. The integer sum at real length differs from the sum at length M by either no term or the term with index $2M+1$. That term has norm at most one. The triangle inequality proves the comparison. Finally $V - 1 \ge V/2$ if $V \ge 2$. □

Lean correspondence: `GuthMaynard.Improvement.RealIntegerSource:source_floor_bounds,source_double_floor,boundedPolynomial_norm_integer_sum,boundedPolynomial_le_detecting_floor_add_one,detecting_floor_large_value`.

1.3 Endpoint claims, with the exact quantifier order

Set

$$A(\sigma)=\begin{cases}3/(2-\sigma), & \sigma\le 9/13,\\ 18/(7\sigma+3), & 9/13<\sigma\le 5/7,\\ 9/4, & 5/7<\sigma\le 3/4,\\ 3/(2\sigma), & 3/4<\sigma,\end{cases}\qquad f(\sigma)=A(\sigma)(1-\sigma).$$

Only the interval $1/2\le\sigma\le 1$ is asserted in the endpoint density results.

Theorem 1.6 (The fixed-line full curve)**.** *For each $\sigma\in[1/2,1]$ and each $\varepsilon>0$ there is $C_{\sigma,\varepsilon}>0$ such that for every real $T\ge 2$,*

$$N(\sigma,T)\le C_{\sigma,\varepsilon}T^{f(\sigma)+\varepsilon}.$$

Theorem 1.7 (The globally uniform coefficient)**.** *For each $\varepsilon>0$ there is $C_\varepsilon>0$ such that simultaneously for $\sigma\in[1/2,1]$ and $T\ge 2$,*

$$N(\sigma,T)\le C_\varepsilon T^{(39/17)(1-\sigma)+\varepsilon}.$$

Theorem 1.8 (Uniform exact curve away from the strict boundary)**.** *For each $\delta>0$ and $\varepsilon>0$ there is $C_{\delta,\varepsilon}>0$ such that for every $\sigma\in[1/2,1]$ satisfying $\sigma\le 3/4$ or $\sigma\ge 3/4+\delta$, and every $T\ge 2$,*

$$N(\sigma,T)\le C_{\delta,\varepsilon}T^{f(\sigma)+\varepsilon}.$$

Theorem 1.9 (Middle-range large values)**.** *For each $\varepsilon>0$ there is $C_\varepsilon>0$ such that for all real N,T,V,t_0, all coefficients $a:\mathbb{N}\to\mathbb{C}$, and all finite $W\subset\mathbb{R}$, if*

$$N\ge 2,\quad N^{6/5}\le T\le N^{5/3},\quad V>0,$$

$|a_n|\le 1$ on I_N, $W\subset[t_0,t_0+T]$, W is one-separated, and $|D_{N,a}(t)|\ge V$ for $t\in W$, then

$$|W|\le C_\varepsilon N^\varepsilon\left(\frac{N^2}{V^2}+\frac{N^{18/7}T^{6/7}}{V^4}\right).$$

Lean correspondence: `GuthMaynardStatement.full_curve,global_uniform,uniform_away_boundary,middle_large_valuesinlean/Solution.lean`.

The proofs of Theorems 1.6–1.8 appear after all branches have been established. Theorem 1.9 is proved in Chapter 5.

Chapter 2

Fourier-affine machinery and its uniform interfaces

2.1 The exact affine average

Definition 2.1. For $R, K \geq 1$ and $C_0 \geq 0$, let

$$\mathcal{B}(R, K, C_0) = \{(b, q, d) \in \mathbb{Z}^3 : R \leq |b| \leq 2R,\ K \leq q \leq 2K,\ |d| \leq C_0 R\}.$$

The slope has both signs; the denominator is positive. Put

$$J_{R,K,C_0}(g) = \int_{\mathbb{R}} \left(\sum_{(b,q,d) \in \mathcal{B}(R,K,C_0)} g\left(\frac{bu+d}{q}\right) \right)^2 du.$$

For a finite subset S the corresponding energy $J_S(g)$ uses S in the sum. For a smooth compactly supported input, let $\|g\|_\infty = \sup_x |g(x)|$. A fixed Fourier profile $D(\beta, n)$ means the family of bounds

$$|\widehat{g}(\xi)| \leq D(\beta, n)\mathcal{T}^\beta(\mathcal{T}/|\xi|)^n \|g\|_\infty \quad (\beta > 0,\ n \in \mathbb{N},\ \xi \neq 0). \tag{2.1}$$

Lean correspondence: `GuthMaynard.Improvement.StrengthenedAffine:sourceSup,FourierSupInput,affineAverage;AffineBox:InAffineBox;AffineFiniteCover:affineIndexIntegral`.

Established analytic input 2.2 (The strengthened unequal-scale affine bound)**.** Fix $0 < a \leq b$, $C_0 \geq 0$, and a nonnegative Fourier profile D. For each $\varepsilon > 0$ there is $C > 0$ such that for all $\mathcal{T} \geq 2$, $1 \leq K \leq R \leq \mathcal{T}^4$, and all nonnegative C^∞ functions g supported in $[a, b]$ satisfying (2.1),

$$J_{R,K,C_0}(g) \leq C\mathcal{T}^\varepsilon \left(R^2K^4 \left(\int g \right)^2 + RK^3 \int g^2 \right). \tag{2.2}$$

The constant precedes the height, both real scales, and the function. There is also a finite-subset and mixed-box version: slope and translation may have separate shells bounded by R, and the denominator shell is bounded by K, with the same right-hand expression up to the fixed support constants.

Lean correspondence: `GuthMaynard.Improvement.StrengthenedAffine.strengthened_unequal_affine;UnequalFullInput.unequal_affine_full_finite;MixedHeightTheorem.mixed_height_affine;MixedMaskedAffine.masked_mixed_affine`.

How this input is proved in the development. The following explanation retains the actual two-stage reduction; it does not replace this input by an unproved affine conjecture. For the source shells M_1, M_2, M_3, the native affine transform integrates the square of $\sum g((m_1u+m_3)/m_2)$; the first two shells are signed nonzero dyadic integer shells and the third is a central integer shell. Its energy $\mathcal{J}(g,M)$ is the finite maximum over $1 \le M_j \le M$. The Fourier-height family is mass-normalized:

$$\sup_{\xi} |\xi|^n |\widehat{g}(\xi)| \le 2^d D(\beta,n) \mathcal{T}^{\beta} \mathcal{T}^{n+2} \int g$$

at smoothing depth d. The support envelope at that depth is $[1/4 - 2d/\mathcal{T},\ 9/4 + 2d/\mathcal{T}]$. The factor $\mathcal{T}^{n+2}$ is not a height-independent constant.

The native Lemma-9.2 reduction supplies, after choosing its Fourier order, a recurrence of the form

$$\mathcal{J}(g,M) \le A\mathcal{T}^{\alpha} M^6 \Big(\int g\Big)^2 + B\mathcal{T}^{\alpha} M^2 \|g\|_2 \sqrt{\mathcal{J}(\widetilde{g}, M_2)} + E\Big(\int g\Big)^2, \qquad M_2 \le M. \tag{2.3}$$

Here A, B, E are uniform functions of the selected Fourier order, divisor constant, and current profile. The adaptive smoothing is the actual `gmAffineTildeSchwartz`; its parameter equals $\mathcal{T} + M_2/(2M_3)$ after division by the first Poisson window. It preserves nonnegativity, enlarges the support by at most $2/\mathcal{T}$, and updates the Fourier coefficient by a factor two. The source proves

$$M_2^6 \Big(\int \widetilde{g}\Big)^2 + M_2^4 \int \widetilde{g}^2 \le 16 \left(M^6 \Big(\int g\Big)^2 + M^4 \int g^2 \right).$$

Let the latter parenthesis be W. For a desired loss δ, choose finitely many descent steps r with $(3/2)^r \delta \ge 100$. At the terminal loss ≥ 100, the crude source bound $\mathcal{J}(g,M) \le 36992 M^6 \int g^2$ and $M \le \mathcal{T}^4$ suffice. At an induction step use (2.3) with $\alpha = \delta/8$ and the induction hypothesis with loss $3\delta/2$ on $\widetilde{g}$. It gives

$$\sqrt{\mathcal{J}(\widetilde{g}, M_2)} \le 4\sqrt{C_{\mathrm{next}}}\, \mathcal{T}^{3\delta/4} \sqrt{W}.$$

Since $M^2\|g\|_2 \le \sqrt{W}$, the mixed term is at most $4B\sqrt{C_{\mathrm{next}}}\mathcal{T}^{7\delta/8}W$. The first and last terms are at most $(A+E)\mathcal{T}^{\delta}W$. Thus $C = A + E + 4B\sqrt{C_{\mathrm{next}}} + 1$ closes the induction. A threshold $\mathcal{T} \ge \max(2, 8r)$ guarantees all support envelopes used by this finite descent. This is why the complete profile must be fixed before the varying function and height.

Lean correspondence: `RiemannZeta.GuthMaynard.exists_uniform_gmAffineJ_le_height_finite_descentandgmAffine_proposition9_1_height_family_nativeinGuthMaynard/AffineSource/LargeValuesAffineHeight.lean`.

The unequal-scale Poisson reduction retains an outer factor RK, not a factor R^2. With $m = \int g$ and $n = \|g\|_2$ it supplies

$$J \le E_0(R^2K^4m^2 + RKn\sqrt{Q}), \qquad Q \le H_0(K^6m^2 + K^4n^2), \quad H_0 \ge 1.$$

Hence $\sqrt{Q} \le \sqrt{H_0}(K^3m + K^2n)$. The cross term RK^4mn is at most $R^2K^4m^2 + RK^3n^2$, by expanding a nonnegative square and using $K \le R$. Therefore

$$J \le 2E_0\sqrt{H_0}(R^2K^4m^2 + RK^3n^2).$$

The source takes $E_0 = A_0\mathcal{T}^{\varepsilon/2}$ and $H_0 = (B_0+1)\mathcal{T}^{\varepsilon}$, so the total loss is exactly $\mathcal{T}^{\varepsilon}$. Heights below 8 are treated by the crude finite bound and $R \le 8^4$, not discarded.

Lean correspondence: `GuthMaynard.Improvement.UniformReduction.unequal_reduction_of_estimates;UnequalHeightTheorem.unequal_height_affine`.

Exact Poisson normalizations. The identities used in that reduction are

$$(\psi(\cdot/R)\widehat{e(\alpha\cdot)})(\xi) = R\widehat{\psi}(R(\xi-\alpha)), \qquad \sum_{d\in\mathbb{Z}} \psi(d/R)e(\alpha d) = R\sum_{\ell\in\mathbb{Z}} \widehat{\psi}(R(\ell-\alpha)).$$

The second Poisson scale is S/K, so its argument is $(S/K)(j - q'u' + qu)$ and its prefactor is S/K. The affine Fourier factor for positive b, q is $e(\xi d/b)(q/b)\widehat{g}(q\xi/b)$. Signed slopes are handled by explicit reindexing. All interchanges require the source's integrability and rapid-decay proofs. Passing from canonical support to general fixed positive support uses $g_1(x) = g(\ell(x-1))$ and the injective index map $(b, q, d) \mapsto (b, \ell q, d+\ell q)$; there is no Jacobian in u.

Lean correspondence: `GuthMaynard.Improvement.PoissonReduction;UnequalFullInput.supportIndexMap_term`.

2.2 Masks and smoothing without regularity of mask boundaries

Let b_κ be the source's `ContDiffBump` at zero with inner radius 1 and outer radius 2. Its normalized kernel is the exact function $\kappa(x) = b_\kappa(x)/\int b_\kappa$; the denominator is strictly positive. Thus κ is nonnegative, compactly supported, smooth and has mass one. Set $\kappa_h(x) = h^{-1}\kappa(x/h)$ and $S_h g = \kappa_h * g$, for $h > 0$.

Lean correspondence: `GuthMaynard.Improvement.CompactKernel:kernelBump,compactKernel,compactKernel_mass`.

Lemma 2.3 (Mass and square norm under smoothing)**.** *For nonnegative integrable g with g^2 integrable,*

$$\int S_h g = \int g, \qquad \int (S_h g)^2 \le \int g^2.$$

If g is supported in $[a, b]$, $S_h g$ is smooth and supported in $[a - 2h, b + 2h]$.

Proof. Tonelli and the mass-one identity give preservation of mass. Apply the weighted Cauchy–Schwarz inequality at each x to $\int \kappa_h(y)g(x-y)\,dy$, then integrate and use translation invariance. The support inclusion follows because $x - y$ cannot be in $[a, b]$ when x is outside its $2h$ enlargement and $|y| \le 2h$. Derivatives may be taken on the compactly supported smooth convolution factor, as justified by the source's convolution regularity theorem. □

Lean correspondence: `GuthMaynard.Improvement.Smoothing:smooth_mass,smooth_l2_contraction;MaskedAffine.compact_smoothed_general_schwartz`.

Proposition 2.4 (The masked affine interface)**.** *Fix $0 < a \le b$ and $C_0 \ge 0$. For each $\varepsilon > 0$ one constant works for all admissible $\mathcal{T}, R, K, h$, every measurable set E, and every measurable real r supported in $[a, b]$ with r^2, r^4 integrable:*

$$J_{R,K,C_0}(S_h(\mathbf{1}_E r^2)) \le C\mathcal{T}^\varepsilon \left(R^2K^4\Big(\int_E r^2\Big)^2 + RK^3 \int_E r^4 \right),$$

where $\mathcal{T} \ge 2$, $1 \le K \le R \le \mathcal{T}^4$, and $0 < h \le \min(1, a/4)$, $h^{-1} \le \mathcal{T}$.

Proof. The smoothed function is nonnegative, smooth, and supported in $[a/2, b + 2]$. The established normalized smoothing theorem supplies (2.1) with constants independent of $E, r, h, \mathcal{T}$; importantly the normalization is by the actual output supremum, not by an inverse mask mass. Apply Input 2.2 on the enlarged fixed interval. The preceding lemma preserves $\int_E r^2$ and bounds the output square integral by $\int_E r^4$. This proves the assertion including the case of zero restricted mass. No regularity of ∂E is used. □

Lean correspondence: `GuthMaynard.Improvement.MaskedAffine:compact_smoothed_full_input,compact_smoothed_unequal_affine,masked_unequal_affine`.

Chapter 3

The centered cubic, exceptional Fourier mass, and spectral comparison

3.1 Literal kernels, measures, and operators

Fix the actual comparison bump w: it is the smooth bump centered at $5/2$ with inner radius $3/2$ and outer radius 2. Thus $0 \le w \le 1$, $w = 1$ on $[1, 4]$, and its closed support is contained in $[1/2, 5]$. Put $a_0(x) = w(x)^2$ and $\chi(s) = w(s + 2)$. Then $0 \le \chi \le 1$, $\chi = 1$ on $[-1, 2]$, and $\operatorname{supp} \chi \subset [-2, 3]$.

Lean correspondence: `MathCollab.Density.ComparisonCutoff;GuthMaynard.Improvement.FixedCutoffs`.

For a smooth amplitude a supported in $[l, u] \subset (0, \infty)$ define

$$K_{a,L}(s) = \sum_{\lceil lL \rceil \le m \le \lfloor uL \rfloor} a(m/L) \mathrm{e}^{\mathrm{i}s \log(m/L)} - L \int_{\mathbb{R}} a(x) \mathrm{e}^{\mathrm{i}s \log x}\, dx. \tag{3.1}$$

For a measurable nonnegative integrable time density μ, set $d\nu_\mu(t) = \mu(t)\, dt$ and

$$\mathcal{C}_{a,L}(\mu) = \iiint K_{a,L}(t_1 - t_2) K_{a,L}(t_2 - t_3) K_{a,L}(t_3 - t_1) \prod_{j=1}^{3} d\nu_\mu(t_j).$$

This is the independent definition `kernelCubic`, not a trace definition. In $\mathcal{H} = L^2(\nu_\mu; \mathbb{C})$, let $p_x(t) = e(t\lambda(x))$. For a real nonnegative cutoff a, define

$$G = \sum_{\lceil lL \rceil \le m \le \lfloor uL \rfloor} a(m/L)\, p_{m/L} \otimes p_{m/L}, \qquad C = L \int a(x)\, p_x \otimes p_x\, dx, \qquad H = G - C.$$

Here $(v \otimes v)z = \langle v, z \rangle v$. These are positive operators G, C, while H need not be positive. The source constructs the integral as a nuclear integral of rank-one operators; $\int \|La(x)p_x\| \|p_x\|\, dx < \infty$ follows from $a \in L^1$ and $\|p_x\|^2 = \nu_\mu(\mathbb{R})$.

Lean correspondence: `GuthMaynard.Improvement.OperatorKernel:operatorDiscrete,operatorContinuous;OperatorCubic;CubicKernel;CenteredPoisson`.

Lemma 3.1 (The genuine cubic trace identity)**.** *For a countable Hilbert basis b of $\mathcal{H}$,*

$$\operatorname{tr}_b(H^3) = \mathcal{C}_{a,L}(\mu), \qquad \operatorname{tr}_b(T) = \sum_j \langle b_j, T b_j \rangle$$

for the nuclear operators used here.

Proof. Expand the nuclear presentations of the three factors. A rank-one cyclic product has trace the cyclic product of three feature inner products. Since $\langle p_x, p_y\rangle = \widehat{\mu}(\lambda(x) - \lambda(y))$ with the source's orientation, integration over frequencies gives the Fourier-side cyclic product; Fubini gives the independently defined time-side kernel cubic. The discrete-continuous difference is represented by the disjoint union of finite counting frequencies and continuous frequencies, with signed amplitude. Absolute integrability of the products is bounded by the product of the nuclear norm integrals. This justifies the expansion and every interchange. The source proves basis-independence through these integral presentations, rather than presuming an infinite eigenbasis. □

Lean correspondence: `GuthMaynard.Improvement.OperatorCubic.operator_frequency_cubic;OperatorCentered.operator_centered_cubic;OperatorTrace`.

3.2 The masked cubic estimate and why the first mask is retained

For the remainder of this section $\mu \geq 0$ is measurable, belongs to $L^1 \cap L^2$, and is supported in an interval of length at most $\mathcal{T}$. Write

$$F = \widehat{\mu}, \quad M = \int \mu^2, \quad M_1 = \int \mu, \quad E = \int |F|^4, \quad E_0 = \int_{B^c} |F|^4, \quad A_2 = \int_B |F|^2.$$

The Fourier square and fourth powers are integrable: Plancherel gives $\int |F|^2 = M$, and $|F| \leq M_1$ gives $\int |F|^4 \leq M_1^2 M$.

Established analytic input 3.2 (Complete masked centered-cubic estimate)**.** For fixed smooth a supported in a fixed positive interval, each $\varepsilon > 0$ has a constant $C > 0$ such that if $\mathcal{T} \geq 2$, $\mathcal{T}^{4/5} \leq L \leq \mathcal{T}$, and B is measurable and symmetric under $\xi \mapsto -\xi$, then

$$|\mathcal{C}_{a,L}(\mu)| \leq C\mathcal{T}^{\varepsilon}\left(\mathcal{T}^2 M^{3/2} + \mathcal{T}L\sqrt{M}\sqrt{E_0} + \sqrt{A_2}(\mathcal{T}^2 M + \mathcal{T}L\sqrt{E})\right). \tag{3.2}$$

The constant precedes all densities, masks, and scales.

Lean correspondence: `GuthMaynard.Improvement.CenteredRestrictedCubic.centered_masked_cubic_estimate`.

Precise analytic construction behind this input. Poisson summation turns (3.1) into the sum of the nonzero modes $L\widehat{(a(x)e^{is\log x})}(Lm)$. The source retains $0 < |m| \leq U$, $U = \mathcal{T}^{1+\eta}/L$, choosing $0 < \eta < 4/5$ before all varying data. It bounds the cubic replacement error by $C\mathcal{T}^{-100}M^{3/2}$. Expanding the retained cube gives the exact factor L^3 and, for $m = (m_1, m_2, m_3)$, the spatial integral with amplitude $a(x_1)a(x_2)a(x_3)$, phase $e(-L(m_1x_1 + m_2x_2 + m_3x_3))$, and Fourier product

$$F(\lambda(x_3) - \lambda(x_1))F(\lambda(x_1) - \lambda(x_2))F(\lambda(x_2) - \lambda(x_3)).$$

Set $x_1 = rv$, $x_2 = ry$, $x_3 = r$. The Jacobian is exactly r^2. The Fourier product becomes $F(-\lambda(v))F(\lambda(v)-\lambda(y))F(\lambda(y))$. Radial integration, before taking a norm, yields rapid decay in $L(m_1v + m_2y + m_3)$. The near region is the thin strip $|L(m_1v + m_2y + m_3)| \leq \mathcal{T}^{\eta}$. It is rewritten as a window of width $\mathcal{T}^{\eta}/(L|m_2|)$ about $-(m_1v+m_3)/m_2$. Negative denominators are normalized by simultaneous negation of modes. For the inverse-ratio energy, the exact identity

$$\frac{-(m_1v + m_3)/m_2}{v} = -\frac{m_3/v + m_1}{m_2}$$

swaps slope and translation. The Jacobian for $v \mapsto 1/v$ is bounded on the fixed positive ratio interval. Both energies are bounded by the mixed masked affine theorem of Chapter 2. The dyadic mode sum, width losses, mode count, and radial discard are charged in the source `cubic_scale_closure` chain. In Fourier coordinates the relevant measure change is $dx = 2\pi x\, d\lambda$; no unrecorded normalization change is made.

For a symmetric exceptional set B, the retained cubic is partitioned exactly into four designated-mask terms

$$(B^c, B^c, B^c), \qquad (B, \mathbb{R}, \mathbb{R}), \qquad (B, \mathbb{R}, B^c), \qquad (B, B^c, B^c).$$

Cyclic reindexing of both spatial coordinates and the full mode cube moves each exceptional mask to the first Fourier factor. The all-low term costs $\mathcal{T}^{\varepsilon}(\mathcal{T}^2 M^{3/2} + \mathcal{T}L\sqrt{M}\sqrt{E_0})$; each of the other three costs $\mathcal{T}^{\varepsilon}\sqrt{A_2}(\mathcal{T}^2 M + \mathcal{T}L\sqrt{E})$. The homogeneous negligible errors are absorbed by the first term. For bounded heights the source separately bounds the cubic by a constant multiple of $\mathcal{T}^2 M^{3/2}$. Thus eventual Poisson retention is not used to skip the bounded-height range.

Lean correspondence: `GuthMaynard.Improvement.CubicRetainedPartition.retained_cubic_designated_partition;RadialCoordinates;CubicMixedBox;CubicFourierMoments;CubicMaskedEstimate;CubicRetainedMasked;CenteredCubicApprox;CenteredCubicBoundedHeight`.

Explicit affine-to-cubic exponent closure. For fixed ratio support and a fixed factor $q_0 \geq 1$, the smoothing width in a denominator shell K is $h = q_0\mathcal{T}^{\eta}/(LK)$. For $1 \leq K \leq R \leq U = \mathcal{T}^{1+\eta}/L$ one has $h^{-1} \leq \mathcal{T}$ and $h \leq q_0\mathcal{T}^{\eta-4/5}$. Since $\eta < 4/5$, a single eventual height threshold puts every such width in the admissible compact-smoothing range of Proposition 2.4. After the affine estimate and Cauchy–Schwarz, a box contribution at restricted first-factor square mass A and other-factor masses M_D, E_D has scalar factor

$$\sqrt{A}\frac{L^2\mathcal{T}^{\eta}}{K}\sqrt{R^2K^4M_D^2 + RK^3E_D}.$$

The square root is at most $K(RKM_D + \sqrt{RK}\sqrt{E_D})$. Both $RK \leq U^2$ and $\sqrt{RK} \leq U$; hence the factor is at most

$$\sqrt{A}\left(\mathcal{T}^{3\eta}\mathcal{T}^2 M_D + \mathcal{T}^{2\eta}\mathcal{T}L\sqrt{E_D}\right),$$

because $L^2\mathcal{T}^{\eta}U^2 = \mathcal{T}^{2+3\eta}$ and $L^2\mathcal{T}^{\eta}U = \mathcal{T}^{1+2\eta}L$. A Fourier-affine loss $\mathcal{T}^{\alpha}$ contributes $\mathcal{T}^{\alpha/2}$ after the square root. The finite dyadic-class count contributes at most $D_{\delta}\mathcal{T}^{\delta}$, since $U \leq \mathcal{T}$. Choose $\eta = \min(\varepsilon/12, 1/10)$, $\alpha = \varepsilon/4$, $\delta = \varepsilon/4$; then $\delta + \alpha/2 + 3\eta \leq \varepsilon$. Thus both terms fit the displayed masked-cubic budget while the restricted factor $\sqrt{A}$ is retained unchanged. Passing back from ratio coordinates charges only fixed factors bounded using $dx = 2\pi x\, d\lambda$.

Lean correspondence: `GuthMaynard.Improvement.CubicScaleClosure:cubic_smoothing_width_bounds,cubic_box_polynomial_scale,cubic_box_epsilon_absorption;CubicScaledStrip;CubicFourierMoments`.

Proposition 3.3 (Restricted cubic from the masked estimate)**.** *For the same $a, \mu, \mathcal{T}, L$ and every $\Lambda \geq 0$,*

$$|\mathcal{C}_{a,L}(\mu)| \leq C\mathcal{T}^{\varepsilon}\left(\mathcal{T}^2 M^{3/2} + \mathcal{T}L\Lambda M + \mathcal{T}LM_1\int_{\{|F|>\Lambda\}}|F|^2\right). \tag{3.3}$$

Proof. Take $B = \{|F| > \Lambda\}$. The Fourier transform is continuous, so B is measurable. Because μ is real, $F(-\xi) = \overline{F(\xi)}$, so B is symmetric. Put $A_2 = \int_B |F|^2$. Plancherel and the pointwise bounds give

$$A_2 \leq M, \qquad E_0 \leq \Lambda^2 M, \qquad E \leq \Lambda^2 M + M_1^2 A_2.$$

It follows that $\mathcal{T}L\sqrt{M}\sqrt{E_0} \le \mathcal{T}L\Lambda M$ and $\mathcal{T}^2M\sqrt{A_2} \le \mathcal{T}^2M^{3/2}$. Also $\sqrt{E} \le \Lambda\sqrt{M} + M_1\sqrt{A_2}$; hence

$$\mathcal{T}L\sqrt{A_2}\sqrt{E} \le \mathcal{T}L\Lambda M + \mathcal{T}LM_1A_2.$$

Substituting these three inequalities in (3.2) increases its constant by at most a factor three. This is precisely the source's restricted closure, including zero mass and zero threshold. □

Lean correspondence: `GuthMaynard.Improvement.CubicOrdinaryClosure.ordinary_fourier_threshold_budgets,ordinary_restricted_cubic_of_masked;CenteredRestrictedCubic.centered_restricted_cubic`.

3.3 The amplifier and the spectral inequality

The amplifier column set is

$$Q_{N,L} = \{q \in \mathbb{N} : \lceil L/N \rceil_+ \le q \le \lfloor 2L/N \rfloor_+\}.$$

For $N > 0$, $N \le L$, its cardinality k satisfies $L/(2N) \le k \le 2L/N$. Its divisor matrix is

$$B_{m,q} = \sum_{n \in I_N} \mathbf{1}_{m=qn} a_n.$$

Every contributing product qn has $1 \le qn/L \le 4$, the plateau of w. Its synthesized column is exactly $\mathrm{e}^{\mathrm{i}t\log(qN/L)}D_{N,a}(t)$. The source bounds the amplifier norm by divisor multiplicity, using the established subpower divisor bound $d(n) \ll_\eta n^\eta$. It then constructs a finite orthonormal family in its synthesized range, not a postulated infinite eigenfamily.

Lean correspondence: `GuthMaynard.Improvement.AmplifierArithmetic;AmplifierNorm;AmplifierSynthesis;AmplifierQuantitative`.

Established analytic input 3.4 (The actual finite amplifier witness)**.** For each $\eta > 0$ there is $P > 0$ such that, uniformly in the bounded coefficients, scales $N > 0$, $N \le L$, and finite nonnegative time measures, there is an orthonormal family $v_1, \dots, v_r$, $r \le k$, satisfying

$$k\frac{\int \mu|D_{N,a}|^2}{P(1+L)^\eta(N+1)} \le \sum_{j=1}^{r}\langle v_j, Gv_j\rangle.$$

Here G uses amplitude w^2 on the indicated integer rows. The source also proves the Schur bound $\|C\| \le D_0LB$ whenever $0 \le \mu \le B$, with D_0 depending only on the fixed cutoff.

Lean correspondence: `GuthMaynard.Improvement.AmplifierQuantitative.amplifier_time_density_witness;OperatorSchur.operator_fixed_cutoff_schur`.

Proof of the finite amplifier witness. Let $\mathcal{A}$ be the literal divisor matrix between the finite column and row Euclidean spaces. For a row m, the product fiber is $F_m = \{(q,n) \in Q_{N,L} \times I_N : qn = m\}$. Cauchy–Schwarz and $|a_n| \le 1$ give $|\mathcal{A}z(m)|^2 \le |F_m| \sum_{(q,n)\in F_m} |z_q|^2$. A nonempty fiber injects into the divisors of $|m|$ and has $0 < m \le 4L$. Thus $|F_m| \le C_\eta(1+L)^\eta$. Summing over rows counts each (q,n) at most once and gives $\|\mathcal{A}\|^2 \le P(1+L)^\eta(N+1)$. Let $\mathcal{B}$ synthesize a row coordinate into $w(m/L)p_{m/L}$. Then $\mathcal{B}\mathcal{B}^* = G$ exactly. Every synthesized column $\mathcal{B}\mathcal{A}e_q$ has square norm $\int \mu|D|^2$ by the plateau identity and its unit phase. Choose an orthonormal basis $v_1, \dots, v_r$ of the finite range of $\mathcal{B}\mathcal{A}$; $r \le k$. Parseval in that range and adjoint Parseval in the finite column space give

$$\sum_q \|\mathcal{B}\mathcal{A}e_q\|^2 = \sum_j \|\mathcal{A}^*\mathcal{B}^*v_j\|^2 \le \|\mathcal{A}\|^2 \sum_j \|\mathcal{B}^*v_j\|^2 = \|\mathcal{A}\|^2 \sum_j \langle v_j, Gv_j\rangle.$$

Insert the matrix bound and divide by its positive uniform upper bound. This proves the displayed witness, including rank zero. The Schur estimate is the separate proved continuous-kernel norm input stated above. □

Lean correspondence: `GuthMaynard.Improvement.AmplifierNorm.amplifier_matrix_energy;AmplifierWitness.amplifier_range_witness;AmplifierSynthesis.amplifier_synthesis_factorization`.

Lemma 3.5 (Finite spectral sign control)**.** *If $C \geq 0$ and $G = C + H \geq 0$ in finite dimension, with eigenvalues h_j of H, then*

$$\sum_j (-h_j)_+^3 \leq \operatorname{tr}(CH^2), \qquad \sum_j (h_j)_+^3 \leq |\operatorname{tr}(H^3)| + \operatorname{tr}(CH^2).$$

Proof. In an orthonormal eigenbasis of H put $c_j = \langle b_j, Cb_j\rangle$. Positivity gives $c_j \geq 0$ and $c_j + h_j \geq 0$. If $h_j \leq 0$, $(-h_j)^3 \leq c_j h_j^2$; otherwise the left side is zero. Sum these inequalities. Finally $(h_j)_+^3 = h_j^3 + (-h_j)_+^3$ and $\sum h_j^3 \leq |\sum h_j^3|$. Notice that C is not assumed to commute with H or to be diagonal. □

Lean correspondence: `GuthMaynard.Improvement.FiniteSpectral.finite_negative_trace,finite_positive_trace;OperatorSpectralComparison.operator_eigenbasis_diagonal_nonneg`.

Established analytic input 3.6 (Nuclear spectral comparison)**.** For positive G, C with the source's nuclear presentations, a finite orthonormal witness satisfying $ka \leq \sum_j \langle v_j, Gv_j\rangle$, $r \leq k$, and $\|C\| \leq B$ obeys

$$k(a - B)_+^3 \leq |\operatorname{tr}_b((G - C)^3)| + \operatorname{Re}\operatorname{tr}_b(C(G - C)^2).$$

The countable basis b exists for the actual L^2 time space.

Justification of the imported operator step. In finite dimension the source constructs an eigenbasis of $G - C$. Bessel's inequality bounds the projections of the finite witness. The positive-part cubic inequality follows from the preceding sign-control lemma together with the finite witness's quadratic lower bound. For the actual space take finite-dimensional spans of the witness and finitely many basis vectors, compress both operators, apply the finite inequality, and pass to the directed limit. Nuclear presentations prove convergence of both cubic and mixed diagonal traces. Thus no infinite spectral-basis assumption or unproved compact trace identity is required.

Lean correspondence: `GuthMaynard.Improvement.OperatorSpectralComparison.operator_finite_spectral_comparison;OperatorNuclearSpectral.operator_nuclear_spectral_comparison;OperatorFiniteTrace;OperatorTraceConvergence`.

Established analytic input 3.7 (The mixed source correction)**.** Fix $A \geq 1$ and $p > 0$. For every $\varepsilon > 0$ one constant works whenever $N \geq 2$, $N \leq L$, $0 \leq \mathcal{T} \leq AN^p$, $\mu \in L^1 \cap L^2$ is measurable and nonnegative, supported in an interval of length $\mathcal{T}$, and $\mu \leq |D_{N,a}|^2$:

$$|\operatorname{tr}_b(CH^2)| \leq R_0 N^\varepsilon L^2 N\mathcal{T} \int \mu^2.$$

Lean correspondence: `GuthMaynard.Improvement.OperatorSourceMixed.operator_source_mixed`.

The source discharges this with the following literal row and cycle bounds. For any fixed reflection cutoff v, put $c_v(s) = \int v(x) \mathrm{e}^{\mathrm{i}s\log x} dx$ and $K = K_{v,L}$. The established source-row theorem states, uniformly for u on the time support,

$$\int \mu(t)|K(u - t)|^2 dt \leq R_v N^\varepsilon LN\mathcal{T}.$$

Its proof uses the actual reflected Mellin kernel: near rows use bounded coefficient/product energy, signed shells use Fourier-reflected band estimates, and finite dyadic summation includes negative reflection. It applies to the genuine centered row, not to an unrelated absolute kernel. The continuous correction has kernel Lc_v. The source proves $\psi = |c_v| \in L^1$, bounded, by compact-smooth Mellin decay. By the nuclear mixed trace identity its cycle is

$$\iiint Lc_v(u-v)K(v-t)K(t-u)\mu(u)\mu(v)\mu(t)\,du\,dv\,dt.$$

Use $|K(-s)| = |K(s)|$ and $|K(v-t)K(t-u)| \le (|K(v-t)|^2 + |K(u-t)|^2)/2$. The row bound controls the inner t integral by $R_v N^\varepsilon L N\mathcal{T}$. The remaining double integral of $\psi(u-v)\mu(u)\mu(v)$ is at most $\|\psi\|_1 \int \mu^2$, by the source's L^1–L^2 convolution inequality. This proves the mixed estimate with constant $(1+\|\psi\|_1)R_v$. In the raw spectral application $v = a_0 = w^2$, which is itself an admissible reflection cutoff; hence the displayed G, C, H are exactly the ones in the row estimate.

Lean correspondence: `GuthMaynard.Improvement.ReflectionSourceRow.centered_source_row;MixedKernelRow.kernelMixed_time_row_bound;MixedCutoffEnvelope;OperatorMixedBound;SourceSpectralRaw`.

Proposition 3.8 (Raw spectral mass estimate)**.** *Under the hypotheses of Input 3.7 and also $\mu \le \min(|D_{N,a}|^2, B)$, $B \ge 0$, let $M = \int \mu^2$. For any $\eta, \varepsilon > 0$ the source supplies constants $P, D_0, R_0 > 0$ such that*

$$M^3 \le 4\big(P(1+L)^\eta(N+1)\big)^3 \left((D_0 LB)^3 + \frac{2N}{L}\left(|\mathcal{C}_{a_0,L}(\mu)| + R_0 N^\varepsilon L^2 N\mathcal{T}M \right) \right). \tag{3.4}$$

Proof. Pointwise $\mu^2 \le \mu|D|^2$, with both integrals justified by boundedness of the finite polynomial. The amplifier witness therefore has level at least M/P_1, where $P_1 = P(1+L)^\eta(N+1) > 0$. Use Input 3.6, $k \ge L/(2N)$, the Schur bound $\|C\| \le D_0 LB$, the cubic trace identity, and Input 3.7. Writing $z = (M/P_1 - D_0 LB)_+$ gives

$$\frac{L}{2N} z^3 \le |\mathcal{C}| + R_0 N^\varepsilon L^2 N\mathcal{T}M.$$

Since $M/P_1 \le D_0 LB + z$ and $(x+y)^3 \le 4(x^3+y^3)$ for nonnegative x, y, multiplication by P_1^3 gives exactly (3.4). This proof also works when $M = 0$. □

Lean correspondence: `GuthMaynard.Improvement.SpectralMassAssembly.actual_spectral_mass_bound,spectral_mass_scalar;SourceSpectralRaw.source_spectral_raw`.

Proposition 3.9 (Uniformly absorbed spectral scale)**.** *Fix $A \ge 1$, $p > 0$. For every $\delta > 0$, one constant $S > 0$ works for $N \ge 2$, $N \le L \le AN^p$, $0 \le \mathcal{T} \le AN^p$, and the preceding source densities:*

$$M^3 \le SN^\delta \left((NLB)^3 + \frac{N^4}{L}|\mathcal{C}_{a_0,L}(\mu)| + N^5 L\mathcal{T}M \right). \tag{3.5}$$

Proof. Take $\eta = \delta/(6p)$ and the mixed-source loss $\varepsilon = \delta/2$. Then $3p\eta + \varepsilon = \delta$. From $1 + L \le (1+A)N^p$ and $N + 1 \le 2N$,

$$4\big(P(1+L)^\eta(N+1)\big)^3 N^\varepsilon \le 32P^3(1+A)^{3\eta}N^{\delta+3}.$$

Insert an additional $N^\varepsilon \ge 1$ in the nonnegative terms of (3.4) where necessary. Expanding N^3 into the three terms yields $(NLB)^3$, $2N^4|\mathcal{C}|/L$, and $2R_0N^5L\mathcal{T}M$. A uniform constant $32P^3(1+A)^{3\eta}(D_0^3 + 2 + 2R_0)$ dominates their coefficients. □

Lean correspondence: `GuthMaynard.Improvement.SpectralScaleBudget:spectral_prefactor_budget,spectral_raw_scale_budget,source_spectral_scale`.

Chapter 4

The literal truncation and the optimized fourth moment

4.1 Truncation is not clipping

For $N \geq 2$, $T > 0$, a bounded source $D = D_{N,a}$, and real U, put

$$\beta_T(t) = \chi(t/T), \qquad \mu_U(t) = \mathbf{1}_{\{|D(t)| \leq U\}} \beta_T(t)|D(t)|^2, \qquad r_U(t) = \mathbf{1}_{\{|D(t)| > U\}} \beta_T(t)|D(t)|^2.$$

The measure density is *discarded* above U; it is not $\min(|D|^2, U^2)$. Let

$$M = \int \mu_U^2, \qquad H_{\mathrm{rem}} = \int r_U, \qquad M_1 = \int \mu_U.$$

All these integrals are finite and nonnegative. Indeed D is a finite continuous sum with $|D| \leq N + 1 \leq 2N$ and β_T is bounded and compactly supported. The source proves measurability and integrability of μ_U, μ_U^2, and r_U, support in $[-2T, 3T]$, and

$$0 \leq \mu_U(t) \leq \min(|D(t)|^2, U^2), \qquad \mu_U + r_U = \beta_T|D|^2.$$

For negative U, $\mu_U = 0$; the domination statement remains valid.

Lean correspondence: `GuthMaynard.Improvement.SourceTailConditional.source_density_regular;TruncatedSourceAttachment.fixed_truncated_source_regular`.

4.2 The arithmetic sampling interface and the exceptional Fourier tail

Established analytic input 4.1 (Actual source sampling)**.** For the fixed χ, one $D_0 > 0$ works for all $N, T \geq 2$ and bounded source coefficients. For every complex test h with $h \in L^1 \cap L^2$, $\widehat{h} \in L^1$, and $\operatorname{supp} h \subset \operatorname{supp} \beta_T$,

$$\left| \int |D(t)|^2 h(t)\, dt \right| \leq D_0 N^2 \int |\widehat{h}| + D_0 N \sqrt{T \log(2N)} \left(\int |h|^2 \right)^{1/2}.$$

Lean correspondence: `GuthMaynard.Improvement.SourceDirichletTail.boundedPolynomial_source_sampling;SourceTailConditional.SourceSamplingBound`.

Arithmetic content of this input. Expansion of $|D|^2$ reduces pairing to Fourier samples at $\log(n/m)/(2\pi)$, $n, m \in I_N$. Reduce (n, m) by its greatest common divisor to (a, b); its actual multiplicity is the number of pairs on that primitive rational ray. It is at most $\lfloor 2N \rfloor / \max(a, b)$. Distinct primitive pairs of height at most Y have logarithms separated by at least Y^{-2}: their integer cross-products differ by at least one, and $\log p - \log q \geq 1/p$ for $p \geq q + 1$. At height q there are at most $2q$ positive pairs. Thus squared multiplicities through height Y have sum at most $2\lfloor 2N \rfloor^2(1 + \log Y)$. The source decomposes exact low height plus half-open dyadic height shells, uses disjoint Fourier-sampling neighborhoods and the proved derivative variation estimate, and obtains the displayed sampling bound. In particular it does not discard repeated ratios, the diagonal ratio $(1, 1)$, or the nonintegral endpoint term. More precisely, set $Y = \min(2N, \max(2, \sqrt{T}))$ and choose the least J with $2N \leq 2^J Y$. Then $Y \geq 2$, $Y^2 \leq 2T$, $\sum_{j<J} 2^j Y \leq 4N$ and $\sum_{j<J} (2^j Y)^{-1} \leq 2/Y$. If $Y = 2N$ then $J = 0$; otherwise $Y \geq \sqrt{T}$. For a test supported in $[-cT, cT]$, with $M_h = \int |h|^2$ and $I_h = \int_{-2}^{2} |\widehat{h}|$, the exact low-height sampling cost is at most

$$\sqrt{16N^2 \log(2N)} \sqrt{(2\pi Y^2 + 4\pi cT) M_h}.$$

The established shell-variation bound at height $Q = 2^j Y$ is $16\pi NQ I_h + 8\pi NcTQ^{-1}\sqrt{M_h}$. The geometric sums therefore give high cost at most $64\pi N^2 I_h + 16\pi c(NT/Y)\sqrt{M_h}$, with the second term absent when $J = 0$. Since $NT/Y \leq N\sqrt{T}$ when it is present, and $\log(2N) \geq 1$, the low and high terms fit

$$(64\pi + 8\sqrt{\pi(1 + c)} + 16\pi c) \left(N^2 I_h + N\sqrt{T \log(2N)} \sqrt{M_h} \right).$$

For the fixed time support take $c = 3$ and enlarge the band integral to the full Fourier L^1 integral. The finite expansion of the polynomial pairing then gives Input 4.1, with bounded coefficients applied only inside I_N.

Lean correspondence: `GuthMaynard.Improvement.ReducedRatioGeometry;SourceRatioAssembly;SamplingVariation;SamplingFourier;SourceRatioHeightBudget;SourceRatioBound`.

Proposition 4.2 (Source Fourier mass above the actual removal threshold)**.** *There is a fixed $D_1 > 0$ such that if $\Lambda \geq D_1(N^2 + H_{\mathrm{rem}})$, then*

$$\int_{\{|\widehat{\mu}_U| > \Lambda\}} |\widehat{\mu}_U|^2 \leq D_1 N^2 T \log(2N). \tag{4.1}$$

Proof. Set $E = \{|\widehat{\mu}_U| > \Lambda\}$, $A_2 = \int_E |\widehat{\mu}_U|^2$, and $g = \mathcal{F}^{-1}(\mathbf{1}_E \widehat{\mu}_U)$. Since $\Lambda > 0$,

$$\int_E |\widehat{\mu}_U| \leq A_2/\Lambda, \quad |g(t)| \leq A_2/\Lambda, \quad \int |g|^2 = A_2.$$

The source proves the inverse-Fourier identity $\int \mu_U g = A_2$; reality makes this bilinear identity have the stated orientation, by the symmetric Fourier mask. For $h = \beta_T g$, its Fourier L^1 norm is bounded by JA_2/Λ, $J = \int |\widehat{\chi}|$, and its square norm by A_2. All test regularity follows from the source's smooth cutoff convolution lemmas. The decomposition $\beta_T |D|^2 = \mu_U + r_U$ and the sampling input imply

$$A_2 \leq (D_0 N^2 J + H_{\mathrm{rem}}) A_2/\Lambda + D_0 N \sqrt{T \log(2N)} \sqrt{A_2}.$$

Choose D_1 large enough that $2(D_0 N^2 J + H_{\mathrm{rem}}) \leq \Lambda$ and $4D_0^2 \leq D_1$. Absorb the first term, obtaining $A_2/2 \leq D_0 N \sqrt{T \log(2N)} \sqrt{A_2}$. If $A_2 = 0$ the result is immediate; otherwise divide by $\sqrt{A_2}$ and square. This proves (4.1) without assuming that the removed mass is small. □

Lean correspondence: `GuthMaynard.Improvement.SourceTailConditional.source_tail_of_sampling,source_tail_scalar_absorption;SourceDirichletTail.boundedPolynomial_source_tail`.

Established analytic input 4.3 (Actual second moments and classical level volume)**.** For $N \geq 2$, $T \geq 0$, bounded source coefficients, and one-separated $W \subset [t_0, t_0 + T]$,

$$\sum_{t \in W} |D(t)|^2 \leq 102N(T + N\log(2N)).$$

For $N \leq T$ the source gives the simpler bound $204TN\log(2N)$. For each $\varepsilon > 0$, whenever $N \geq 2$, $N \leq T \leq N^2$, $V > 0$,

$$|W| \leq C_\varepsilon N^\varepsilon \left(N^2/V^2 + TN^4/V^6\right) \tag{4.2}$$

for every one-separated source large-value set. The corresponding Lebesgue level volume on an interval of length qT is at most $qC_\varepsilon N^\varepsilon(N^2/V^2 + TN^4/V^6)$ for integer $q \geq 0$. Finally, for our truncated source with $N \leq T$,

$$M_1 \leq 612TN\log(2N).$$

Lean correspondence: `GuthMaynard.Improvement.SeparatedSecondMoment;ClassicalLargeValues.classical_large_values;ClassicalLevelVolume.boundedPolynomial_classical_level_volume;TruncatedSourceMeanValue.fixed_truncated_source_mass`.

No missing threshold case in the classical estimate. The high-threshold input behind (4.2) is the native integer large-values theorem on $T^{1/3} \leq M \leq T$, with threshold $V \geq M^{3/4+\kappa}$, giving $T^\eta(M^2/V^2 + M^3\sqrt{T}/V^4)$. The real-to-integer transfer of Chapter 1 applies above $16N^{3/4+\varepsilon/8}$; nonempty W implies $N \geq 8$, which gives the integer scale condition. Its second term fits (4.2), since

$$N^3\sqrt{T}/V^4 \leq N^2/V^2 + TN^4/V^6$$

by expanding $(NV^2 - N^2\sqrt{T})^2 \geq 0$. Below the threshold the separated second moment is used, with $V^4 \leq 16^4 N^{3+\varepsilon/2}$ and $\log(2N) \leq (\log 2 + 2/\varepsilon)N^{\varepsilon/2}$. Empty sets are handled directly. Thus the classical input is not applied outside its large-value height range.

Lean correspondence: `GuthMaynard.Improvement.ClassicalLargeValues:classical_cross_budget,classical_low_scalar,classical_floor_scale,classical_large_values`.

Proposition 4.4 (Removed mass bound)**.** *For each $\varepsilon > 0$ one constant works for $N \geq 2$, $N \leq T \leq N^2$, $U \geq 1$, and every bounded coefficient family:*

$$H_{\mathrm{rem}} \leq C_\varepsilon N^\varepsilon(N^2 + TN^4/U^4). \tag{4.3}$$

Proof. Choose a dyadic cutoff J with $2N \leq 2^J$ and $J+1 \leq K_\varepsilon N^{\varepsilon/2}$. Such a cutoff is obtained from the least dyadic cover and the logarithmic power bound. Since $|D| \leq 2N \leq 2^J U$, split the removed part into levels $2^j U < |D| \leq 2^{j+1}U$, $0 \leq j < J$. On each level $|D|^2 \leq 4(2^jU)^2$. The cutoff is at most one and supported in $[-2T, 3T]$, an interval of length $5T$, so Input 4.3 with loss $\varepsilon/2$ gives

$$H_{\mathrm{rem}} \leq 5CN^{\varepsilon/2} \sum_{j<J} 4(2^jU)^2 \left(N^2/(2^jU)^2 + TN^4/(2^jU)^6\right).$$

The first part is $4JN^2$. The second is $4TN^4U^{-4}\sum_{j<J} 16^{-j} \leq 8TN^4U^{-4}$. Use the bound on $J+1$ and $N^{\varepsilon/2} \geq 1$; the product of the two half-losses is N^ε. The source chooses $40CK$ as a common constant. □

Lean correspondence: `GuthMaynard.Improvement.RemovedMassBound:removed_dyadic_classical_sum,removed_dyadic_cover,fixed_source_removed_mass`.

4.3 Fixed-source raw cubic and spectral assembly

Use Proposition 3.3 on support length $5T$, with $\Lambda = D_1(N^2 + H_{\mathrm{rem}})$. By (4.1) and $M_1 \le 612TN\log(2N)$,

$$|\mathcal{C}_{a_0,L}(\mu_U)| \le C(5T)^{\varepsilon}\left(25T^2M^{3/2} + 5TLD_1(N^2+H_{\mathrm{rem}})M + 3060D_1N^3LT^3\log^2(2N)\right). \tag{4.4}$$

This requires $(5T)^{4/5} \le L \le 5T$, not merely $T^{4/5} \le L$. The coefficient 3060 is the product $5 \cdot 612$, and the support translation is $[-2T, 3T]$.

Lean correspondence: `GuthMaynard.Improvement.TruncatedSourceCubic.fixed_truncated_source_cubic`.

If also $N \le T \le N^2$, $N \le L$, then $L, 5T \le 5N^2$. Apply Proposition 3.9 with $A = 5$, $p = 2$, $B = U^2$. Substituting (4.4) gives

$$M^3 \le SN^{\delta}\Bigg[(NLU^2)^3 + 5N^5LTM \tag{4.5}$$

$$+ \frac{N^4}{L}C(5T)^{\varepsilon}\Big(25T^2M^{3/2} + 5TLD_1(N^2+H_{\mathrm{rem}})M \tag{4.6}$$

$$+ 3060D_1N^3LT^3\log^2(2N)\Big)\Bigg].$$

The constants S, C, D_1 precede all sources and scales.

Lean correspondence: `GuthMaynard.Improvement.TruncatedSourceMomentRaw.fixed_truncated_source_moment_raw`.

4.4 The three optimized quantities

Definition 4.5. For $N, T > 0$ define

$$L_* = N^{1/7}T^{5/7}, \qquad U_* = N^{5/7}T^{1/14}, \qquad P_* = N^{18/7}T^{6/7}.$$

Lemma 4.6 (Admissibility and exact scale identities)**.** *If $N \ge 2$ and $N^{6/5} \le T \le N^{5/3}$, then*

$$N \le T \le N^2, \quad N \le L_*, \quad T^{4/5} \le L_* \le T, \quad 1 \le U_* \le N,$$

and $5L_*$ is admissible for (4.5)*. Moreover*

$$NL_*U_*^2 = P_*, \qquad N^4T^2/L_* = P_*^{3/2} = P_*\sqrt{P_*},$$
$$N^5L_*T = P_*^2, \qquad TN^4/U_*^2 = P_*,$$
$$TN^4/U_*^4 = NL_*, \qquad N^7T^3 \le P_*^3.$$

Proof. All powers have positive bases. $N \le T \le N^2$ follows from $1 \le 6/5$ and $5/3 \le 2$. From $T \ge N^{6/5}$, $T^{5/7} \ge N^{6/7}$, giving $N \le L_*$. From $T \le N^{5/3}$, $T^{3/35} \le N^{1/7}$, giving $T^{4/5} = T^{3/35}T^{5/7} \le L_*$. Using $N \le T$, $L_* \le T^{6/7} \le T$. Multiplying by five and $5^{4/5} \le 5$ yields $(5T)^{4/5} \le 5L_* \le 5T$. Also $U_* \ge 1$ and, using $T \le N^2$, $U_* \le N^{6/7} \le N$. Every identity is an exact addition of exponents; for example $NL_*U_*^2$ has N-exponent $1+1/7+10/7 = 18/7$ and T-exponent $5/7 + 1/7 = 6/7$. For the final inequality, $N^7T^3/P_*^3 = N^{-5/7}T^{3/7} \le 1$ because $T^3 \le N^5$. □

Lean correspondence: `GuthMaynard.Improvement.MiddleRangeScales;MiddleRangeIdentities;MiddleLargeValueBranches.middle_threshold_le_length`.

Lemma 4.7 (Homogeneous cubic absorption)**.** *If $M, P, Q \geq 0$ and*

$$M^3 \leq Q(P^3 + P^{3/2}M^{3/2} + P^2M),$$

then $M \leq (1+3Q)P$. If the right-hand coefficient is QN^ε with $N \geq 1$, $\varepsilon \geq 0$, then $M \leq (1+3Q)N^\varepsilon P$.

Proof. If $M \leq P$, the first conclusion is immediate. Otherwise $M > 0$ and $P \leq M$. Each term in the bracket is at most PM^2: $P^3 \leq PM^2$, $P^{3/2}M^{3/2} \leq PM^2$, and $P^2M \leq PM^2$. Thus $M^3 \leq 3QPM^2$; cancellation of $M^2 > 0$ gives $M \leq 3QP$. For the second assertion use the first with QN^ε and then $1 + 3QN^\varepsilon \leq (1+3Q)N^\varepsilon$. The argument does not divide by P, so it includes $P = 0$. □

Lean correspondence: `GuthMaynard.Improvement.MomentScalarClosure.normalized_moment_absorption,normalized_moment_power_absorption`.

Theorem 4.8 (Actual optimized truncated fourth moment)**.** *For every $\varepsilon > 0$ there is $C_\varepsilon > 0$ such that, uniformly for $N \geq 2$, $N^{6/5} \leq T \leq N^{5/3}$, and bounded coefficients,*

$$\int_{\mathbb{R}} \left(\mathbf{1}_{\{|D_{N,a}| \leq U_*\}} \chi(t/T)|D_{N,a}(t)|^2\right)^2 dt \leq C_\varepsilon N^\varepsilon P_*.$$

Proof. Write $L = L_*$, $U = U_*$, $P = P_*$ and $\theta = \varepsilon/6$. Use (4.5) with $\delta = \theta$, cubic loss θ, and spectral scale $5L$. Lemma 4.6 turns it exactly into

$$M^3 \leq SN^\theta \Big[125P^3 + 25P^2M + C(5T)^\theta \Big(5P^{3/2}M^{3/2} + 5D_1N^4T(N^2 + H_{\mathrm{rem}})M + 3060D_1N^7T^3\log^2(2N)\Big)\Big]. \tag{4.7}$$

The 125 and 25 are the literal factor-five rescaling losses. Use Proposition 4.4 with loss θ, giving

$$H_{\mathrm{rem}} \leq KN^\theta(N^2 + NL) \leq 2KN^\theta NL, \qquad N^2 + H_{\mathrm{rem}} \leq (1+2K)N^\theta NL.$$

The latter and $N^5LT = P^2$ bound the middle removal term by $5D_1(1+2K)N^\theta P^2M$. Moreover

$$(5T)^\theta \leq 5^\theta N^{2\theta}, \qquad \log^2(2N) \leq J^2N^{2\theta}, \quad J = \log 2 + 1/\theta, \qquad N^7T^3 \leq P^3.$$

Since $N^\theta, N^{2\theta} \geq 1$, all nonnegative terms in (4.7) are dominated by

$$QN^{\theta+\theta+2\theta+2\theta}(P^3 + P^{3/2}M^{3/2} + P^2M),$$

where the source's explicit uniform coefficient is

$$Q = S\left(150 + C5^\theta\left(5 + 5D_1(1+2K) + 3060D_1J^2\right)\right) \geq 0.$$

The four losses sum to $6\theta = \varepsilon$. Lemma 4.7 gives $M \leq (1+3Q)N^\varepsilon P$. Every constant was selected before N, T, a. □

Lean correspondence: `GuthMaynard.Improvement.MiddleMomentRaw.middle_source_moment_raw;MomentScalarClosure.raw_moment_normalized_budget;MiddleFourthMoment.middle_fourth_moment`.

Chapter 5

From the fourth moment to all separated large values

5.1 Good cells, bad cells, and the reproducing weight

The source fixes the exact frequency cutoff χ of Chapter 3 and defines $k = \mathcal{F}^{-1}\chi$ as a Schwartz function. Put $q(t) = |k(t)|$; no smoothness of the absolute value q is required. For $n \in I_N$ the negative frequency $-\log(n/N)/(2\pi)$ lies in $[-1, 2]$, so the transform of k equals one there. Translation invariance gives, term by term,

$$\int k(t-w)e(t\xi)\,dt = e(w\xi)\widehat{k}(-\xi).$$

Finite summation therefore proves $D(w) = \int k(t-w)D(t)\,dt$. This applies inversion to the Schwartz kernel, not to the nonintegrable exponential. Two weighted Cauchy–Schwarz inequalities give

$$|D(w)|^4 \le \left(\int q\right)^3 \int q(t-w)|D(t)|^4 dt.$$

Schwartz decay supplies $q(s) \le C_j(1+|s|)^{-j}$. Applying the source's separated quadratic-decay packing lemma gives the uniform overlap bound $\sum_{w\in W} q(t-w) \le B_0$. Since $\int |s|^j q(s)\,ds$ is finite, $q(s) \le |s|^j q(s)/R^j$ on $|s| > R$. Multiply by $|D|^4 \le 16N^4$ and integrate after translating. This proves, for $R > 0$,

$$\int_{|t-w|>R} q(t-w)|D(t)|^4 dt \le D_j N^4/R^j.$$

All weighted moments are integrable because D is bounded and the kernel is integrable; no global L^4 assertion for D is used.

Lean correspondence: `GuthMaynard.Improvement.SourceReproducingKernel;SourceQuarticKernel;SourceQuarticTail`.

Let

$$\mathcal{K} = \{k \in \mathbb{Z} : \lfloor -T\rfloor \le k \le \lfloor 2T\rfloor,\ \exists x \in [-T, 2T],\ |D(x)| > U,\ \lfloor x\rfloor = k\}.$$

The exact deleted set is $Z = \bigcup_{k\in\mathcal{K}}\{w \in W : |w-k| \le R+1\}$, and $G = W \setminus Z$. A one-separated set in $[k-R-1, k+R+1]$ has at most $2R+3$ points. Taking the union bound proves $|Z| \le (2R+3)|\mathcal{K}|$. If $w \in G$ and $|t-w| \le R \le T$, then $t \in [-T, 2T]$. If $|D(t)| > U$, the integer $k = \lfloor t\rfloor$ belongs to $\mathcal{K}$, and $|t-k| < 1$ implies $|w-k| \le R+1$, contradicting $w \notin Z$. Thus $|D(t)| \le U$ and $\chi(t/T) = 1$, so $\mu_U(t)^2 = |D(t)|^4$. This explains the plateau $[-1, 2]$.

For completeness, the bad-cell bound uses parity-bin selection alone. Choose one witness from each occupied cell and select the parity class containing at least half the cells. Those witnesses are one-separated. Split them into the three intervals $[-T, 0]$, $[0, T]$, $[T, 2T]$; each has length T, preserving $T \le N^2$. The classical bound applied at threshold U gives at most three copies of its budget. Restoring both parities gives at most six copies. No derivative estimate or positive lower bound on the measure of a bad cell is needed.

Lean correspondence: `GuthMaynard.Improvement.QuarticCellGeometry;QuarticMomentExtraction.quartic_good_integral`.

Proposition 5.1 (Extraction at a specified truncated moment)**.** *For each $\varepsilon > 0$ there is $E_\varepsilon > 0$ such that if $N \ge 2$, $N \le T \le N^2$, $1 \le V \le U \le 2N$, and $W \subset [0, T]$ is one-separated with $|D(w)| \ge V$, then for every upper bound M_0 on $\int \mu_U^2$,*

$$|W| \le E_\varepsilon N^\varepsilon \left(N^2/U^2 + TN^4/U^6 + M_0/V^4 \right). \tag{5.1}$$

Proof. For a fixed tail order j, the preceding reproducing estimates imply for $w \in G$

$$V^4 \le A_0 \int q(t-w)\mu_U(t)^2 dt + A_0 D_j N^4/R^j, \qquad A_0 = \left(\int q \right)^3.$$

When the last term is at most $V^4/2$, sum over G, interchange the finite sum and integrals, and use the overlap bound to obtain $|G| \ll M_0/V^4$. The actual classical bad-cell bound gives $|\mathcal{K}| \ll_\varepsilon N^\varepsilon(N^2/U^2 + TN^4/U^6)$. It uses occupied unit cells and the parity argument just proved, not the measure of a superlevel set or an uncharged derivative-variation loss. Combining deleted and good points gives

$$|W| \le B \left((R+1)N^{\varepsilon/2}(N^2/U^2 + TN^4/U^6) + M_0/V^4 \right)$$

whenever its uniform rapid-tail condition holds.

Choose $\nu = \min(\varepsilon/2, 1/2) > 0$ and an integer j such that $\nu j \ge 5$. Let $N_0 = \max(2, 2B)$ and take $R = N^\nu$ for $N \ge N_0$. Then $R \le N \le T$, $R^j \ge N^5$, and $BN^4/R^j \le 1/2 \le V^4/2$. Also $(R+1)N^{\varepsilon/2} \le 2N^\varepsilon$. For $2 \le N < N_0$, packing gives $|W| \le T + 1 \le 2N_0^2$, while $U \le 2N$ gives $N^2/U^2 \ge 1/4$. Thus an enlarged common constant $8N_0^2 + 2B + 1$ covers the bounded-length case too. This proves (5.1) with no radius or tail premise left. □

Lean correspondence: `GuthMaynard.Improvement.QuarticMomentExtraction.quartic_moment_extraction;QuarticBadCells.quartic_bad_cells_bound;QuarticExtraction.quartic_extraction;QuarticRadiusExtraction.quartic_extraction_of_moment`.

Lemma 5.2 (Threshold budgets)**.** *In the middle time range, if $0 < V \le U_*$, then*

$$N^2/U_*^2 + TN^4/U_*^6 \le 2P_*/V^4.$$

If $V \ge U_*$, then $TN^4/V^6 \le P_*/V^4$.

Proof. Since $N \le L_*$ and $P_* = NL_*U_*^2$, $N^2/U_*^2 \le P_*/U_*^4$. The identity $TN^4/U_*^2 = P_*$ gives $TN^4/U_*^6 = P_*/U_*^4$. Sum and use $V^4 \le U_*^4$. For the second assertion use $V^2 \ge U_*^2$ in $TN^4/V^6 = (TN^4/V^2)/V^4$. □

Lean correspondence: `GuthMaynard.Improvement.MiddleLowLargeValues.middle_low_threshold_budget;MiddleLargeValueBranches.middle_high_threshold_budget`.

Proof of Theorem 1.9. First treat $W \subset [0,T]$. If $0 < V < 1$, no polynomial estimate is needed: packing gives $|W| \le T+1 \le 2N^2 \le 2N^2/V^2$. If $1 \le V \le U_*$, apply Proposition 5.1 with loss $\varepsilon/2$ and Theorem 4.8 with loss $\varepsilon/2$. The threshold-budget lemma gives

$$|W| \le EN^{\varepsilon/2}\left(2P_*/V^4 + DN^{\varepsilon/2}P_*/V^4\right) \le E(2+D)N^{\varepsilon}P_*/V^4.$$

If $V > U_*$, apply the actual classical estimate (4.2) with loss ε and the high-threshold budget. This gives $|W| \le C_2N^{\varepsilon}(N^2/V^2 + P_*/V^4)$. The sum $C = C_1 + C_2 + 2$ of the low, high, and packing constants works for all positive thresholds.

For a translated interval set $W_0 = \{t - t_0 : t \in W\}$ and $b_n = a_n \mathrm{e}^{\mathrm{i}t_0 \log(n/N)}$. Translation preserves cardinality and separation, $|b_n| = |a_n|$, and

$$D_{N,b}(s) = D_{N,a}(s + t_0).$$

Apply the origin bound to W_0, b. The right side and constant are unchanged. □

Lean correspondence: `GuthMaynard.Improvement.MiddleLowLargeValues.middle_large_values_low;MiddleLargeValueBranches.middle_large_values_high,middle_large_values_small;SourceTranslation;MiddleLargeValues.middle_range_large_values`.

Chapter 6

Short zero detection and fixed coefficient families

6.1 The arithmetic mollifier and the exact contour identity

Let $\mu_{\mathrm{ar}}(n)$ be the integer Möbius function, extended by zero at $n = 0$. The mollifier and its product coefficients are

$$M_X(s) = \sum_{1 \le d \le \lfloor X \rfloor_+} \mu_{\mathrm{ar}}(d) d^{-s}, \qquad c_X(n) = \sum_{d \mid n,\ d \le \lfloor X \rfloor_+} \mu_{\mathrm{ar}}(d).$$

The divisor set at 0 is empty. For $X \ge 1$, $c_X(1) = 1$ and $c_X(n) = 0$ if $2 \le n \le X$. Also $|c_X(n)| \le d(n)$. Define, for $Y > 0$,

$$S(\rho; X, Y) = \sum_{n \ge 0} c_X(n) n^{-\rho} \mathrm{e}^{-n/Y}, \qquad R(\rho; X, Y) = M_X(1) Y^{1-\rho} \Gamma(1-\rho),$$

where the $n = 0$ summand is zero. The source proves absolute convergence in the real-part ranges in which it is used. With $\rho = \beta + \mathrm{i}\gamma$, put

$$I(\rho; X, Y) = \frac{1}{2\pi} \int_{\mathbb{R}} \Gamma(1/2 - \beta + \mathrm{i}v) \zeta(1/2 + \mathrm{i}(\gamma + v)) \cdot M_X(1/2 + \mathrm{i}(\gamma + v)) Y^{1/2 - \beta + \mathrm{i}v} \, dv.$$

Lean correspondence: `MathCollab.Density.MollifierCoefficients;DetectorMellin.detectorKernel;DetectorContour.detectorResidue;DetectorError.detectorIntegral`.

Established analytic input 6.1 (Actual short contour identity)**.** If $\zeta(\rho) = 0$, $9/13 \le \beta < 1$, and $Y \ge 1$, then, for independent mollifier and smoothing parameters X, Y,

$$S(\rho; X, Y) = R(\rho; X, Y) + I(\rho; X, Y).$$

What the contour argument uses. Mellin inversion of $\mathrm{e}^{-n/Y}$ gives the right vertical contour on $\operatorname{Re} s = 1/2$. Move to $\operatorname{Re} s = 1/2 - \beta$ in a finite rectangle. The factor $\zeta(\rho) = 0$ cancels the Gamma singularity at $s = 0$; the zeta pole at $s = 1 - \rho$ gives exactly R. The shifted zeta regularization, residue identity, absolute integrability of both vertical contours, and decay of both horizontal contours are proved dependencies. Taking the height to infinity gives the displayed identity with factor $1/(2\pi)$, not $1/(2\pi\mathrm{i})$ after real vertical parameterization. No Weyl estimate is an extra premise of this identity.

Lean correspondence: `GuthMaynard.Improvement.ShortDetectorIdentity.short_smoothedDetector_contour_identity,short_zeta_detector_identity;ShortDetectorContour;ShortDetectorMellin;ShortDetectorEstimates`.

6.2 The finite detector and its dichotomy

For a positive height H choose

$$X_H = 2H^{1/100}, \quad Y_H = H^{1/2}, \quad K_H = \left\lceil H^{1/2}(\log H)^2 \right\rceil_+ .$$

Define fixed coefficients

$$b_H(n) = \begin{cases} c_{X_H}(n)\mathrm{e}^{-n/Y_H}, & \lfloor X_H \rfloor_+ < n < K_H, \\ 0, & \text{otherwise}, \end{cases} \qquad F_H(\rho) = \sum_{n<K_H} b_H(n) n^{-\rho}.$$

They depend on H, not on either coordinate of the zero. For $H \geq 1$, $K_H > 1$, and $\beta \geq 0$, the source proves the exact split

$$S(\rho; X_H, Y_H) = \mathrm{e}^{-1/Y_H} + F_H(\rho) + \sum_{n\geq 0} c_{X_H}(n + K_H)(n + K_H)^{-\rho}\mathrm{e}^{-(n+K_H)/Y_H}. \tag{6.1}$$

The finite source interval is strict at both X_H and $Y_H(\log H)^2$, because the upper index is $n < K_H$.

Lean correspondence: `GuthMaynard.Improvement.ShortFiniteDetector:shortDetectorCoefficient,shortFiniteDetector,shortFiniteDetector_decomposition`.

Established analytic input 6.2 (Uniform short-detector remainder bounds)**.** For every fixed $c > 0$, eventually in H, the infinite tail in (6.1) has norm at most c uniformly for $\beta \geq 0$; the residue $R(\rho; X_H, Y_H)$ has norm at most c uniformly for $9/13 \leq \beta \leq 1$ and $H \leq |\gamma| \leq 2H$.

These estimates use the exponential smoothing tail and actual Gamma-strip bounds. In particular, the negative-strip bound keeps its singular real-part factor explicit:

$$|\Gamma(-a + \mathrm{i}v)| \leq (12/a)(1 + |v|)\mathrm{e}^{-|v|} \quad (0 < a \leq 1/2).$$

Here $a = \beta - 1/2 \geq 5/26$ for the short branch. The source obtains this negative-strip bound from $\Gamma(z+1) = z\Gamma(z)$ and $|z| \geq a$.

Lean correspondence: `GuthMaynard.Improvement.ShortDetectorTruncation.short_detector_tail_eventually_small;ShortDetectorResidue.short_detector_residue_eventually_small;ShortDetectorGamma.short_detector_gamma_bound`.

Lemma 6.3 (Uniform detector alternative)**.** *Eventually in H, every zero ρ with $9/13 \leq \operatorname{Re}\rho < 1$, $H \leq |\operatorname{Im}\rho| \leq 2H$ satisfies*

$$|F_H(\rho)| \geq 1/3 \quad \textit{or} \quad |I(\rho; X_H, Y_H)| \geq 1/3.$$

Proof. Take both remainders in Input 6.2 to be at most $1/24$, and $Y_H \geq 4$. Since $\mathrm{e}^{-1/Y_H} \geq 1 - 1/Y_H \geq 3/4$, the contour identity and (6.1) give $3/4 \leq |F_H(\rho)| + |I(\rho; X_H, Y_H)| + 1/12$. If both displayed detector values were less than $1/3$, the right side would be less than $2/3 + 1/12 = 3/4$, a contradiction. All required thresholds precede ρ. □

Lean correspondence: `GuthMaynard.Improvement.ShortDetectorAlternative.short_detector_eventually_alternative`.

6.3 Eliminating the varying real part by an exact Taylor family

Put $D_H = \lfloor \log_2 K_H \rfloor + 1$ in the natural-logarithm-to-base-two integer convention `Nat.log`. Decompose the finite detector into blocks $(Q, 2Q]$, $Q = 2^r$, $0 \le r < D_H$:

$$B_{H,Q}(\rho) = \sum_{Q<n\le 2Q} b_H(n) n^{-\rho}.$$

If $|F_H(\rho)| \ge 1/3$, a block has norm at least $1/(3D_H)$. Every nonzero block has $H^{1/100} < Q < H^{1/2}(\log H)^2$. For $j \ge 0$ define

$$b_{H,Q,j}(n) = b_H(n)(2n/Q - 3)^j, \qquad P_{H,Q,j}(t) = \sum_{Q<n\le 2Q} b_{H,Q,j}(n) \mathrm{e}^{-\mathrm{i}t\log(n/Q)}.$$

Again the coefficients do not depend on β or γ. Let

$$w_j(\beta) = \frac{\beta(\beta+1)\cdots(\beta+j-1)}{j!}(-1/3)^j, \qquad w_0(\beta) = 1.$$

For $0 \le \beta \le 1$ the rising-product ratio lies in $[0,1]$, so $|w_j(\beta)| \le 3^{-j}$. The binomial series gives the exact identity

$$\sum_{j\ge 0} w_j(\beta) P_{H,Q,j}(\gamma) = (3/2)^\beta \sum_{Q<n\le 2Q} b_H(n)(n/Q)^{-\beta} \mathrm{e}^{-\mathrm{i}\gamma\log(n/Q)}.$$

Indeed $1 + (2n/Q - 3)/3 = 2n/(3Q)$. The norm of the right side is $(3/2)^\beta Q^\beta |B_{H,Q}(\rho)|$.

Lean correspondence: `MathCollab.Density.DetectorTaylorFamily.detectorTaylorWeight;GuthMaynard.Improvement.ShortDetectorTaylorFamily;ShortDetectorBlockExpansion.hasSum_shortDetectorTaylorPolynomial`.

Proposition 6.4 (A fixed logarithmic-size detecting family)**.** *Let $J_H = \lceil 10 \log H \rceil_+$. Eventually in H, the fixed family (r, j), $0 \le r < D_H$, $0 \le j \le J_H$, has size at most $60(\log H)^2$. If $0 \le \sigma \le \beta \le 1$ and $|F_H(\rho)| \ge 1/3$, one member with $Q = 2^r$ satisfies*

$$|P_{H,Q,j}(\gamma)| \ge Q^\sigma/(60 \log H), \qquad H^{1/100} < Q < H^{1/2}(\log H)^2.$$

Proof. On $(Q, 2Q]$, $|2n/Q - 3| \le 1$, $|b_H(n)| \le d(n) \le n \le 2Q$; therefore $|P_{H,Q,j}(t)| \le 2Q^2$ for every j, t. The infinite Taylor tail after J has norm at most $3Q^2 3^{-J}$. At $J = J_H + 1$, $3^{-J} \le H^{-5}$, and $Q \le H^2$ makes it at most $3/H$. Eventually $3/H \le 1/(6D_H)$. For the block of norm $\ge 1/(3D_H)$ the full Taylor combination has norm at least $Q^\beta/(3D_H)$. Removing the tail leaves at least $Q^\beta/(6D_H)$. The total absolute Taylor weight is at most $\sum 3^{-j} = 3/2 \le 2$, so one retained polynomial has norm at least $Q^\beta/(12D_H)$. Use $Q^\sigma \le Q^\beta$ and $D_H \le 5 \log H$. Finally $J_H + 1 \le 12 \log H$, giving the stated family cardinality. No coefficient is chosen after seeing the zero. □

Lean correspondence: `GuthMaynard.Improvement.ShortFixedDetectorFamily.shortFiniteDetector_fixed_family_eventually;ShortDetectorFamilyScales.short_detector_family_scales_eventually`.

6.4 Powering, dyadic product pieces, and normalization

Extend $b_{H,Q,j}$ by zero outside $(Q, 2Q]$ to an arithmetic function. Its k-fold Dirichlet convolution has coefficients

$$c_{H,Q,j,k}(n) = \sum_{n_1 \cdots n_k = n} \prod_{i=1}^{k} b_{H,Q,j}(n_i).$$

It vanishes unless $Q^k < n \le (2Q)^k$ for $k \ge 1$. The finite Dirichlet series is exactly the kth power of the original series. Use conjugated coefficients $\overline{c_{H,Q,j,k}(n)}$ so that positive phases give the same norm at the *same* ordinate γ. Splitting $(Q^k, 2^k Q^k]$ into k dyadic intervals with starts $N = 2^\ell Q^k$, $0 \le \ell < k$, one interval has polynomial norm at least $|P_{H,Q,j}(\gamma)|^k/k$.

Lean correspondence: `GuthMaynard.Improvement.ShortDetectorConvolution;ShortDetectorPoweredPieces.exists_shortPoweredDetector_piece`.

Lemma 6.5 (Least-successful powering)**.** *Uniformly for $a \in [3/4, 4/5]$ and large H, if $H^{1/100} < Q \le H^{1/2}(\log H)^2$, there is $1 \le k \le 100$ with $H^a \le Q^k \le H^{3a/2}$. Every product piece satisfies*

$$H^a \le N \le 2^{99} H^{3a/2}, \qquad 2N \le H^2.$$

Proof. Since $Q^{100} > H \ge H^a$, choose the least successful k. If $k \le 2$, $Q^k \le Q^2 \le H(\log H)^4 \le H^{3a/2}$ eventually, uniformly because $3a/2 \ge 9/8 > 1$. If $k \ge 3$, leastness gives $Q^{k-1} < H^a$; $k/(k-1) \le 3/2$ gives $Q^k < H^{3a/2}$. For the dyadic pieces use $1 \le 2^\ell \le 2^{99}$. The uniform buffer $2^{100} H^{3a/2} \le H^2$ follows from $3a/2 \le 6/5 < 2$. □

Lean correspondence: `GuthMaynard.Improvement.ShortDetectorPowerScales.short_detector_power_exists,short_detector_power_buffer_eventually,short_detector_power_piece_bounds`.

For fixed $k \le 100$, divisor growth gives $|c_{H,Q,j,k}(n)| \ll_\alpha n^\alpha$, uniformly in H, Q, j, k. For any $\eta > 0$, when $n \le H^2$, eventually these coefficients are at most H^η. Define the normalized coefficients

$$a_{H,Q,j,k,\eta}(n) = \overline{c_{H,Q,j,k}(n)}/H^\eta.$$

They have modulus at most one on all eligible pieces. The full index set is

$$\mathcal{P}_H = \{(r, j, k, \ell) : 0 \le r < D_H,\ 0 \le j \le J_H, 1 \le k \le 100,\ 0 \le \ell < 100\}.$$

Its scale is $N_p = 2^\ell (2^r)^k$ and its coefficient is the preceding normalized function. Including extra pieces with $\ell \ge k$ in this fixed index set is harmless; the actual covering witness has $\ell < k$.

Proposition 6.6 (The powered fixed-family cover)**.** *For each $\eta > 0$, eventually in H, $|\mathcal{P}_H| \le 600000(\log H)^2$. For $a \in [3/4, 4/5]$, $0 \le \sigma \le \operatorname{Re}\rho \le 1$, and $|F_H(\rho)| \ge 1/3$, there is $p \in \mathcal{P}_H$ such that*

$$H^a \le N_p \le 2^{99} H^{3a/2}, \qquad 2N_p \le H^2, \qquad |D_{N_p, a_{p,\eta,H}}(\operatorname{Im}\rho)| \ge N_p^\sigma H^{-2\eta}.$$

Proof. Use Proposition 6.4, the least-successful power, and the piece pigeonhole. The factors $(60 \log H)^k$, k, and the fixed conversion from $Q^{k\sigma}$ to $(2^\ell Q^k)^\sigma$ are bounded by a constant times $(\log H)^{100}$ uniformly for $k \le 100$, $0 \le \sigma \le 1$. For each fixed $\eta > 0$, the source proves that this entire height loss is at most H^η eventually. Division by the coefficient normalization H^η gives $N_p^\sigma H^{-2\eta}$. The family cardinality is at most $100 \cdot 100$ times $60(\log H)^2$. The coefficient bound is uniform over all family members with $2N_p \le H^2$. This construction preserves the ordinate, avoiding an uncharged reseparation loss. □

Lean correspondence: `GuthMaynard.Improvement.ShortPoweredDetectorFamily.shortPoweredDetectorFamily_cover,shortPoweredDetectorFamily_coefficients;ShortDetectorPoweredHeight;ShortDetectorNormalization`.

Chapter 7

The two density windows from the improved large-values estimate

7.1 The variable short-window exponents

For $9/13 \le \sigma \le 5/7$, define

$$a(\sigma) = \frac{6}{7\sigma+3}, \qquad b(\sigma) = \frac{3}{2}a(\sigma), \qquad g(\sigma) = \frac{18(1-\sigma)}{7\sigma+3}.$$

The source proves

$$3/4 \le a \le 4/5, \quad 0 \le b \le 6/5, \quad g > 0, \qquad 2b(1-\sigma) = g, \qquad a(18/7-4\sigma)+6/7 = g.$$

Also

$$11/6 - 5\sigma/3 \le g(\sigma). \tag{7.1}$$

For completeness, multiplying the difference in (7.1) by $6(7\sigma+3) > 0$ gives $70\sigma^2-155\sigma+75 = 70(\sigma-5/7)(\sigma-3/2) \ge 0$.

Lean correspondence: `GuthMaynard.Improvement.ShortDensityScale.short_density_scale_bounds`, `short_density_scale_identities`.

Proposition 7.1 (Large values in the variable powered window)**.** *For $0 < \delta \le 1/100$ there is a positive constant independent of σ such that, for $H, N \ge 2$, $9/13 \le \sigma \le 5/7$,*

$$H^{a(\sigma)} \le N \le H^{b(\sigma)+\delta}, \qquad V = N^{\sigma}H^{-2\delta},$$

and any bounded source, one-separated $W \subset [t_0, t_0+H]$ with $|D(w)| \ge V$ obeys

$$|W| \le C_\delta H^{g(\sigma)+10\delta}.$$

Proof. We have $N^\delta \le H^{2\delta}$ and $N^{2-2\sigma} \le H^{g+2\delta}$. If $N \le H^{5/6}$, then $N^{6/5} \le H$. The lower bound $N \ge H^a \ge H^{3/5}$ gives $H \le N^{5/3}$. Thus Theorem 1.9 is used in its exact time range. Its two terms, after division by the threshold, are

$$N^\delta\left(N^{2-2\sigma}H^{4\delta} + N^{18/7-4\sigma}H^{6/7+8\delta}\right).$$

Since $18/7-4\sigma \le 0$, the lower length bound and the scale identity give $N^{18/7-4\sigma}H^{6/7} \le H^g$. The first term costs at most $H^{g+6\delta}$ inside the parentheses; the second at most $H^{g+8\delta}$. The outer N^δ costs $H^{2\delta}$, proving this branch with $2CH^{g+10\delta}$.

If $N > H^{5/6}$, use the actual separated second moment. Division by V^2 gives

$$|W| \le 102\left(HN^{1-2\sigma} + N^{2-2\sigma}\log(2N)\right)H^{4\delta}.$$

Because $1-2\sigma \le 0$, $HN^{1-2\sigma} \le H^{11/6-5\sigma/3} \le H^g$. With $D_\delta = \log 2 + 1/\delta$, the logarithm is at most $D_\delta N^\delta \le D_\delta H^{2\delta}$, so the second term costs at most $D_\delta H^{g+4\delta}$. This branch costs at most $102(1+D_\delta)H^{g+8\delta}$ and hence the claimed bound. The source chooses $2C + 102(1+D_\delta)$ as a common constant. □

Lean correspondence: `GuthMaynard.Improvement.ShortWindowLargeValues.short_window_large_values;ShortDensityHeight.short_density_height_budgets`.

7.2 The closed upper window and the padded time

For $5/7 \le \sigma \le 3/4$ set

$$h(\sigma) = \frac{9}{4}(1-\sigma), \qquad s(\sigma) = \begin{cases} 6/7, & \sigma \le 18/25, \\ 7/8, & \sigma > 18/25. \end{cases}$$

The exact scalar comparisons are

$$39/14 - 3\sigma \le h, \quad s(18/5 - 4\sigma) \le h, \quad 1 + s(1-2\sigma) \le h.$$

The first is equality at $5/7$. For the $6/7$ split the respective latter differences are $(-117+165\sigma)/140$ and $(11-15\sigma)/28$; for the $7/8$ split they are $(-18+25\sigma)/20$ and $(3-4\sigma)/8$. They are nonnegative on the specified subintervals.

Lean correspondence: `GuthMaynard.Improvement.UpperCurveBudgets.upper_curve_exponents`.

Proposition 7.2 (Large values in the closed upper powered window)**.** *For $0 < \delta \le 1/100$ there is one constant for all $5/7 \le \sigma \le 3/4$, $H, N \ge 2$, and $H^{3/4} \le N \le H^{9/8+\delta}$, with the same bounded coefficients, separation, localization, and threshold $V = N^\sigma H^{-2\delta}$:*

$$|W| \le C_\delta H^{h(\sigma)+10\delta}.$$

Proof. As before $N^\delta \le H^{2\delta}$ and $N^{2-2\sigma} \le H^{h+2\delta}$. If $N \le H^{s(\sigma)}$, use the padded time

$$S = \begin{cases} H, & N \le H^{5/6}, \\ N^{6/5}, & N > H^{5/6}. \end{cases}$$

It satisfies $H \le S$ and $N^{6/5} \le S \le N^{5/3}$. When $S = H$, the lower bound $N \ge H^{3/4}$ gives $N^{18/7-4\sigma}S^{6/7} \le H^{39/14-3\sigma} \le H^h$. When $S = N^{6/5}$, the term is $N^{18/5-4\sigma}$; its exponent is nonnegative, so the upper length bound gives at most $H^{s(18/5-4\sigma)} \le H^h$. Apply Theorem 1.9 on the enlarged interval $[t_0, t_0+S]$. The original set and threshold are unchanged. The same two-term calculation as in Proposition 7.1 gives at most $2CH^{h+10\delta}$. For $N > H^{s(\sigma)}$, the separated second moment and the last scalar comparison give $HN^{1-2\sigma} \le H^h$; the logarithmic and first-term budgets give the same mean-value closure. This proof includes $\sigma = 3/4$. □

Lean correspondence: `GuthMaynard.Improvement.UpperCurveBudgets.upper_curve_padded_time,upper_curve_mean_budget;UpperCurveLargeValues.upper_curve_large_values`.

7.3 Type-I separated zero representatives

Proposition 7.3 (Type-I counts in both windows)**.** *For each $\varepsilon > 0$, uniformly in σ, t_0 and eventually in H, any finite $R \subset \mathbb{C}$ whose ordinates are injective and one-separated, with $\sigma \le \operatorname{Re}\rho \le 1$, $\operatorname{Im}\rho \in [t_0, t_0 + H]$ and $|F_H(\rho)| \ge 1/3$, has*

$$|R| \le C_\varepsilon H^{g(\sigma)+\varepsilon} \quad (9/13 \le \sigma \le 5/7), \qquad |R| \le C_\varepsilon H^{h(\sigma)+\varepsilon} \quad (5/7 \le \sigma \le 3/4).$$

Proof. Take $\eta = \min(\varepsilon/100, 1/100)$. For the short window apply Proposition 6.6 with $a = a(\sigma)$; for the upper window use $a = 3/4$. Restrict the fixed family to members satisfying the appropriate length bounds and $2N_p \le H^2$. Eventually $2^{99} \le H^\eta$, so its upper length becomes $H^{b+\eta}$ or $H^{9/8+\eta}$ respectively. For each member let W_p be the subset of ordinates where that member has norm at least $N_p^\sigma H^{-2\eta}$. The corresponding window proposition bounds $|W_p| \le C_\eta H^{g+10\eta}$ or $C_\eta H^{h+10\eta}$. Each W_p is a subset of the original separated set, so separation and localization are preserved. The cover gives $W \subset \bigcup_p W_p$. Eventually $600000(\log H)^2 \le H^\eta$. Hence $|R| = |W| \le \sum_p |W_p| \le C_\eta H^{g+11\eta}$, or the version with h. Since $11\eta \le \varepsilon$, the desired estimate follows. All finite families and their coefficients were constructed before any zero or real part was chosen. □

Lean correspondence: `GuthMaynard.Improvement.ShortTypeICount.short_typeI_separated_count`, `short_typeI_zero_slab_count`; `UpperCurveTypeICount.upper_curve_typeI_separated_count`, `upper_curve_typeI_zero_slab_count`.

7.4 The actually used Type-II moment is weaker than the named contract

Put

$$Z_H(t) = |\zeta(1/2 + \mathrm{i}t) M_{2H^{1/100}}(1/2 + \mathrm{i}t)| .$$

The proposition `ShortMobiusFourthMoment` is the assertion $\int_{H/2}^{3H} Z_H^4 \ll_\varepsilon H^{1+\varepsilon}$. Some conditional intermediate modules explicitly take that assertion as a hypothesis. *The final short and upper-curve branches do not use it.* They instead use the following proved weaker theorem.

Established analytic input 7.4 (Native ordinary fourth moment)**.** For every $\varepsilon > 0$ there are $B > 0$ and a height threshold such that

$$\int_x^{2x} |\zeta(1/2 + \mathrm{i}t)|^4 dt \le B x^{1+\varepsilon}$$

for all positive x above that threshold.

This is the native actual-zeta producer `Fourth.zeta_fourth_dyadic`, not a postulated moment inequality. The following expands the actual producer above its already established contour and Gamma-ratio machinery.

Lean correspondence: `MathCollab.Density.Stronger.Fourth.FourthMoment.zeta_fourth_dyadic`.

Exact divisor-square source. Let $s_t = 1/2 + \mathrm{i}t$, $d(n)$ be the divisor count ($d(0) = 0$), $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, $G_t = \Gamma_{\mathbb{R}}(s_t)\Gamma_{\mathbb{R}}(s_{-t}) \ne 0$, and $A(w) = (1 - 4w^2)^4$. Define

$$P_t(w) = \left(\frac{(s_{-t} + w)(1 - s_{-t} - w)}{s_{-t}(1 - s_{-t})} \right)^2 ,$$

$$K_t(w) = \frac{\mathrm{e}^{100w^2} A(w) P_t(w) \Gamma_{\mathbb{R}}(s_{-t}+w)^2}{G_t w}.$$

For $c > 0$ put

$$C_{t,c}(n) = \frac{1}{2\pi} \int_{\mathbb{R}} d(n) n^{-s_{-t}-c-\mathrm{i}u} K_t(c+\mathrm{i}u)\, du, \qquad P_{t,c,N} = \sum_{0 \le n \le N} C_{t,c}(n),$$

with the zero-index term explicitly zero. The exact source's right piece R_t is its vertical divisor integral at real coordinate 1 divided by G_t. The native contour reflection and completed-zeta identity give $z(t)^2 = 2\,\mathrm{Re}\, R_t$, hence $z(t)^4 \le 4|R_t|^2$. For $t \ge 0$, each $C_{t,c}(n)$ is independent of $c > 0$: a finite rectangle has zero boundary integral, both vertical lines are integrable, and both horizontal integrals tend to zero. The source proves $R_t = \sum_{n \ge 0} C_{t,c}(n)$ and the exact finite-prefix-plus-tail split. These are the precise contour identities used here, not an assumed approximate functional equation with unnamed errors.

Lean correspondence: `MathCollab.Density.Stronger.Fourth.SquareSource;SquareNormalized;SquareLineShift;SquareTruncation`.

Truncation and Gaussian prefix control. For $b > 3/2$ the established Gamma-kernel bound and $d(n) \le n$ give

$$|R_t - P_{t,c,N}| \le A_b t^b N^{3/2-b}$$

when $t \ge \max(4, 4b)$, for every $c > 0$. The tail sum is bounded by the decreasing-power integral $\sum_{n>N} n^{1/2-b} \le N^{3/2-b}/(b-3/2)$. For any $\delta > 0$ choose $b = 2 + 2/\delta$; then $b + (1+\delta)(3/2-b) = -(1+\delta)/2 \le 0$. If $H \le t \le 2H$ and $N \ge H^{1+\delta}$, this proves a uniform constant bound $|R_t - P_{t,c,N}| \le K_{\mathrm{tail}}$.

For fixed $c > 0$ the kernel estimate is $|K_t(c+\mathrm{i}u)| \le A_c t^c \mathrm{e}^{-98u^2}$ for $t \ge \max(4, 4c)$. Write $a_c(n) = d(n) n^{-1/2-c}$ and $D_{c,N}(v) = \sum_{1 \le n \le N} a_c(n) \mathrm{e}^{-\mathrm{i}v \log n}$. The exact finite-prefix integral and weighted Cauchy–Schwarz give

$$|P_{t,c,N}|^2 \le K_g t^{2c} \int_{\mathbb{R}} \mathrm{e}^{-98u^2} |D_{c,N}(u-t)|^2 du.$$

Both weighted integrals are finite since $D_{c,N}$ is a bounded finite sum, and the Gaussian mass is $\sqrt{\pi/98}$. The divisor subpower estimate gives, uniformly in $c \ge 0$, $|a_c(n)|^2 \le D_\theta n^{\theta-1}$; summing on $(M, 2M]$ gives $\sum |a_c(n)|^2 \le D_\theta (2M)^\theta$. The proved dyadic-prefix mean square, followed by Gaussian Fubini, therefore supplies, for $N = 2^M$,

$$\int_H^{2H} \int_{\mathbb{R}} \mathrm{e}^{-98u^2} |D_{c,N}(u-t)|^2 du\, dt \le D_p (M+1)^2 (H + 2(5\pi+1)N) N^\theta.$$

Lean correspondence: `MathCollab.Density.Stronger.Fourth.SquareTail;SquarePrefix;Coefficients;PrefixMean;GaussianProducer`.

The actual fourth-moment exponent closure. For requested loss $\varepsilon > 0$ the source's expression `mineps1/20` parses as $q = \min(\varepsilon, 1)/20$. In particular $0 < q \le 1/20$, $q \le 1$, and $7q \le \varepsilon$. Choose $H^{1+q} \le N = 2^M \le 2H^{1+q}$. The tail and prefix estimates give

$$z(t)^4 \le 8K_g (2H)^{2q} \int_{\mathbb{R}} \mathrm{e}^{-98u^2} |D_{q,N}(u-t)|^2 du + 8K_{\mathrm{tail}}^2.$$

The dyadic-count estimate $(M+1)^2 \le A_q N^q$ and $H + 2(5\pi+1)N \le (1 + 4(5\pi+1))H^{1+q}$ show the integrated main budget is at most a constant times

$$(2H)^{2q} H^{1+q} (2H^{1+q})^{2q} = 2^{4q} H^{1+5q+2q^2} \le 2^{4q} H^{1+7q}.$$

The integrated constant tail is $8K_{\text{tail}}^2 H$. Since $7q \le \varepsilon$ and $H \ge 1$, both fit $CH^{1+\varepsilon}$. The height threshold is fixed before H; all finite-prefix, Gaussian, and coefficient constants were already fixed by q. This proves exactly Input 7.4, relative to the explicit contour-kernel inputs.

Lean correspondence: `MathCollab.Density.Stronger.Fourth.FourthMoment.zeta_fourth_dyadic;GaussianProducer.exists_fourth_dyadic_budget`.

Proposition 7.5 (Weak short mollified fourth moment)**.** *For every $\varepsilon > 0$, eventually in H,*

$$\int_{H/2}^{3H} Z_H(t)^4 dt \le C_\varepsilon H^{1+1/25+\varepsilon}.$$

Proof. The elementary mollifier bound at $\operatorname{Re} s = 1/2$ is $|M_X(s)| \le \lfloor X \rfloor_+ \le X$. Thus $Z_H(t)^4 \le 16H^{1/25}|\zeta(1/2+\mathrm{i}t)|^4$. Cover $[H/2, 3H]$ by $[H/2, H]$, $[H, 2H]$, and $[2H, 4H]$. For sufficiently large H the native fourth-moment bound applies on all three intervals. Their sum is at most $3(B+1)(2H)^{1+\varepsilon}$. Multiplying by $16H^{1/25}$ gives the source's constant $48(B+1)2^{1+\varepsilon}$. Closed versus half-open integration endpoints have Lebesgue measure zero. □

Lean correspondence: `GuthMaynard.Improvement.WeakShortMoment.shortTwistedNorm_fourth_le,weak_short_moment`.

Established analytic input 7.6 (Window reduction for actual Type-II detectors)**.** There is $C > 0$, eventually in H, uniformly for $\sigma \ge 9/13$, one-separated $W \subset [H, 2H]$, and a chosen zero representative ρ_u with $\operatorname{Im} \rho_u = u$, $\sigma \le \operatorname{Re} \rho_u \le 1$, such that $|I(\rho_u; X_H, Y_H)| \ge 1/3$ implies

$$|W| \le CH^{1-2\sigma} \int_{H/2}^{3H} Z_H(t)^4 dt.$$

Precise reduction. The detector-window theorem, after Gamma-tail truncation and Hölder, supplies for each u a lower bound

$$1 \le AH^{1-2\sigma} \int_{-H/2}^{H/2} |\Gamma(1/2 - \operatorname{Re} \rho_u + \mathrm{i}v)| Z_H(u+v)^4 dv.$$

Sum over the one-separated ordinates. The uniform negative-Gamma strip bound and the proved separated Gamma-packing bound dominate the sum of weights by a constant on $[H/2, 3H]$. This gives the displayed input. Different representatives are allowed to have different real parts; no common-β assumption is added.

Lean correspondence: `GuthMaynard.Improvement.ShortDetectorWindowDetection.short_detector_window_fourth;ShortTypeIIPacking.short_typeII_window_packing;ShortTypeIICount.short_typeII_count_by_moment`.

Proposition 7.7 (The unconditional Type-II count used here)**.** *For every $\varepsilon > 0$, eventually in H, Type-II separated representatives in a positive slab have*

$$|R| \le C_\varepsilon H^{2-2\sigma+1/25+\varepsilon}.$$

On $9/13 \le \sigma \le 5/7$ this exponent without ε is at most $g(\sigma)$; on $5/7 \le \sigma \le 3/4$ it is at most $h(\sigma)$.

Proof. Multiply Input 7.6 by Proposition 7.5. The exponent is $(1-2\sigma) + (1 + 1/25 + \varepsilon)$. For the first comparison multiply $g - (2 - 2\sigma + 1/25)$ by $25(7\sigma + 3) > 0$: the result is $(7\sigma - 5)(50\sigma - 69) \ge 0$ on the short interval. For the second comparison the difference is $(1-\sigma)/4 - 1/25 \ge 1/16 - 1/25 > 0$ on the closed upper interval. Thus the genuine fixed power loss $1/25$ fits both target windows. □

Lean correspondence: `GuthMaynard.Improvement.WeakTypeIICount.weak_short_typeII_count;WeakShortDensity.weak_short_typeII_zero_slab_count,weak_short_typeII_exponent_le;UpperCurveDensity.upper_curve_slab_density`.

Chapter 8

Restoring analytic multiplicities and summing the height slabs

8.1 Local Jensen counts and separated representatives

Established analytic input 8.1 (The precise local analytic bounds)**.** For $\sigma \geq 3/5$, $H \geq 8$, and an occupied ordinate bin $[k, k+1)$, center the disks at $c = 2 + \mathrm{i}(k + 1/2)$. All slab zeros in that bin are inside the closed disk of radius $3/2$. Zeta is analytic on the disk of radius $7/4$, has $|\zeta(c)| \geq 0.6$, and on its outer circle $|\zeta| \leq 100H^3$. Jensen's divisor bound with analytic multiplicity gives

$$\sum_{\substack{\rho \in Z(\sigma;H,2H) \\ k \leq \operatorname{Im}\rho < k+1}} m(\rho) \leq \frac{\log(100H^3/0.6)}{\log((7/4)/(3/2))}.$$

The geometric inclusion follows from $|\operatorname{Re}(\rho - c)| \leq 7/5$, $|\operatorname{Im}(\rho - c)| \leq 1/2$, and $(7/5)^2 + (1/2)^2 < (3/2)^2$. The outer disk avoids the pole 1 at these heights. The lower Euler-product bound, outer zeta-growth bound, and analytic-order/divisor identification are the exact imported analytic facts used by Jensen.

Lean correspondence: `GuthMaynard.Improvement.ShortLocalZeroCount.short_zeta_unit_bin_sum_le_jensen;MathCollab.Density.LocalZeroCount;MathCollab.Density.ZetaGrowth`.

Lemma 8.2 (Uniform logarithmic local multiplicity and extraction)**.** *There is $B > 0$ such that for all $\sigma \geq 9/13$, $H \geq 2$, every unit bin in $Z(\sigma; H, 2H)$ has total multiplicity at most $B \log(2H)$. There is therefore a subset $R \subset Z(\sigma; H, 2H)$ with injective, one-separated ordinates and*

$$n(\sigma; H, 2H) \leq 2B \log(2H)|R|.$$

Proof. For large H, Input 8.1 is at most a fixed multiple of $\log(2H)$. For bounded $2 \leq H \leq H_0$, every bin sum is at most the finite count $Q = N(9/13, 2H_0)$; since $\log(2H) \geq \log 4$, add $Q/\log 4$ to the uniform coefficient.

For the finite weighted extraction, partition the occupied unit bins into even and odd classes and choose the class carrying at least half the total weight. Choose one zero from each occupied bin in that class. Different chosen bins are separated by at least two integer bin indices; their ordinates are at least one apart and cannot coincide. A bin has weight at most $B \log(2H)$, so the total weighted count is at most $2B \log(2H)$ times the number of representatives. The source packages this as weighted separated extraction, preserving full zero multiplicity. □

Lean correspondence: `GuthMaynard.Improvement.ShortLocalZeroCount.short_zeta_local_multiplicity_bound,short_zeta_separated_representatives;MathCollab.Density.ZeroSeparation`.

Proposition 8.3 (Actual slab density in the two closed windows)**.** *For every $\varepsilon > 0$, one constant and one eventual height threshold work for all σ in either specified interval:*

$$n(\sigma; H, 2H) \le C_\varepsilon H^{g(\sigma)+\varepsilon} \quad (9/13 \le \sigma \le 5/7),$$

$$n(\sigma; H, 2H) \le C_\varepsilon H^{h(\sigma)+\varepsilon} \quad (5/7 \le \sigma \le 3/4).$$

Proof. Choose representatives as in Lemma 8.2. Filter them into $R_I = \{\rho : |F_H(\rho)| \ge 1/3\}$ and its complement R_{II}. The detector alternative implies that every complement member has $|I(\rho; X_H, Y_H)| \ge 1/3$. Both subsets preserve separation and ordinate injectivity. Apply Proposition 7.3 with loss $\varepsilon/2$ and Proposition 7.7 with loss $\varepsilon/2$, using its exact exponent comparisons. Then

$$|R| \le (C_I + C_{II}) H^{q(\sigma)+\varepsilon/2}, \qquad q = g \text{ or } h.$$

Multiplicities are restored by the preceding logarithmic factor. Since $\log(2H) \le (\log 2 + 2/\varepsilon) H^{\varepsilon/2}$, the total exponent is $q(\sigma) + \varepsilon$. The constant is selected before σ, H. □

Lean correspondence: `GuthMaynard.Improvement.ShortDetectorCounting.short_slab_count_of_separated_branch_bounds;WeakShortDensity.short_slab_density;UpperCurveDensity.upper_curve_slab_density`.

8.2 From eventual positive slabs to both-sign all-height counts

The source proves conjugation symmetry of zeta and invariance of its analytic multiplicity, hence $N(\sigma, T) \le 2n(\sigma; 0, T)$. The inequality, rather than equality, permits the real-axis overlap. A closed-slab split satisfies $n(\sigma; 0, 2H) \le n(\sigma; 0, H) + n(\sigma; H, 2H)$; zeros on the shared boundary can be counted twice only on the upper-bound side.

Lean correspondence: `MathCollab.Density.ZetaConjugation`.

Lemma 8.4 (Exact dyadic summation with positive exponent)**.** *For fixed σ, if $p > 0$ and eventually $n(\sigma; H, 2H) \le CH^p$, $C > 0$, then one $A > 0$ works for all $T \ge 2$: $N(\sigma, T) \le AT^p$.*

Proof. First extend the slab estimate to every $H \ge 1$. Choose $R = \max(1, H_0)$ from its eventual threshold and put $Q = n(\sigma; 0, 2R)$. For $1 \le H < R$, monotonicity gives the slab count at most $Q \le QH^p$; for $H \ge R$ use CH^p. Thus $B = C + Q$ works on every such slab. Put $r = 2^p > 1$, $Q_0 = n(\sigma; 0, 1)$ and $D = Q_0 + B/(r-1) > 0$. The source notes $D + B \le Dr$. Induction using the closed-slab split gives

$$n(\sigma; 0, 2^j) \le D(2^j)^p.$$

For $T \ge 1$ choose $2^j \le T < 2^{j+1}$. Monotonicity gives $n(\sigma; 0, T) \le D(2^{j+1})^p \le DrT^p$. Conjugation bounds $N(\sigma, T)$ by twice that quantity. The geometric constant $1/(2^p - 1)$ is finite because $p > 0$. No epsilon is spent in this summation; its dependence on σ is retained. □

Lean correspondence: `MathCollab.Density.DensityDyadicSummation:zetaSlab_extend_power_bound,zetaPositive_bound_of_slabs,zetaDensity_bound_of_eventual_slabs`.

Corollary 8.5 (Fixed-line density in the two new windows)**.** *For* $9/13 \le \sigma \le 5/7$*, every* $\varepsilon > 0$ *gives* $N(\sigma, T) \ll_{\sigma,\varepsilon} T^{g(\sigma)+\varepsilon}$ *for all* $T \ge 2$*. For* $5/7 \le \sigma \le 3/4$ *the same assertion holds with* $h(\sigma)$*.*

Proof. The exponents $g(\sigma) + \varepsilon$ and $h(\sigma) + \varepsilon$ are strictly positive on these closed intervals. Apply Lemma 8.4 to Proposition 8.3 for the fixed line. The source does not infer uniformity in σ from this fixed-line geometric summation. □

Lean correspondence: `GuthMaynard.Improvement.WeakShortDensity.short_density;UpperCurveDensity.upper_curve_density`.

Chapter 9

The previously established lower and strictly upper branches

9.1 The native Ingham branch and count-convention bridge

The lower input is not taken from the informal name "Ingham" alone. The development proves a positive-slab theorem through its native classical detector, then translates the native rectangle count into the actual count of Chapter 1.

Definition 9.1 (Classical exponent certificate)**.** The native logarithmic scale is $\tau = \log T/\log Q$ when $Q > 1$. Put

$$E_{\mathrm{MHH}}(\sigma,\tau) = \max(2-2\sigma, \min(\tau+1-2\sigma, \tau+4-6\sigma)),$$

$$E_{\mathrm{sub}}(\sigma,\tau_0,\tau) = \max(2-2\sigma, 3-3\sigma+\tau-\tau_0).$$

The source structure `EndpointScaleCertificate` consists of $\tau_0 > 0$, $3-3\sigma \le \tau_0$, and the following four obligations:

(i) $E_{\mathrm{MHH}} \le 2-2\sigma$ for $0 \le \tau \le \tau_0+\sigma-1$;

(ii) $E_{\mathrm{sub}} \le (3-3\sigma)\tau/\tau_0$ for $2\tau_0/3 \le \tau \le \tau_0$;

(iii) for $4\tau_0/3 \le \tau \le 8\tau_0/3$, either $\tau/2$ or $\tau/3$ is in $[2\tau_0/3, \tau_0]$;

(iv) the exceptional zeta-polynomial interval $2 \le \tau < 4\tau_0/3$ is empty or lies below $6\sigma-3$, the proved Weyl exclusion threshold.

Lean correspondence: `GuthMaynard.NativeMoment.FiniteScaleAssembly:classicalMHHExponent,corollary1110Envelope,EndpointScaleCertificate,ZetaWindowResolved`.

Lemma 9.2 (The Ingham certificate)**.** *For $1/2 < \sigma < 1$, $\tau_0 = 2-\sigma$ satisfies this certificate.*

Proof. $\tau_0 > 0$ and $3-3\sigma \le 2-\sigma$ follow from $\sigma > 1/2$. In (i) the upper bound is 1, and the second-moment branch of E_{MHH} is at most $2-2\sigma$ for $\tau \le 1$. For (ii), the first component fits the target by $\tau \ge 2\tau_0/3$. For the second, multiplying by $\tau_0 > 0$ reduces the comparison to $(\tau_0-(3-3\sigma))(\tau_0-\tau) \ge 0$. For (iii), choose the half-scale if $\tau \le 2\tau_0$, otherwise the third-scale. For (iv), $4(2-\sigma)/3 < 2$ when $\sigma > 1/2$, so the interval is empty. These are exactly the finite exponent obligations; they do not themselves supply the analytic detector estimate. □

Lean correspondence: `GuthMaynard.NativeMoment.FiniteScaleAssembly.ingham_endpoint_scale_certificate`.

Established analytic input 9.3 (Native classical endpoint positive-slab theorem)**.** For $1/2 < \sigma < 1$ and an endpoint certificate at τ_0, the actual native closed-rectangle count satisfies

$$\forall \varepsilon > 0, \qquad n_{\mathrm{nat}}(\sigma; T, 2T) \ll_{\sigma,\varepsilon} T^{3(1-\sigma)/\tau_0+\varepsilon} \quad \text{eventually in } T.$$

At $\sigma = 1/2$, the native global count separately obeys an ε-power bound relative to T.

The analytic content consumed by this input. The native finite detector dichotomy first gives zero count zero, a Type-I finite polynomial witness, or a Type-II witness. The Type-I logarithmic scale is split into the low range, the basic range $[2\tau_0/3, \tau_0]$, the raised range $\tau \geq 4\tau_0/3$, and the remaining medium range. In the latter the source separates $\tau \leq 11/10$ from $11/10 < \tau < 4\tau_0/3$. The low/basic/powered consumers use actual MHH, convolution-power, coefficient normalization, and displacement/reseparation bounds. The last medium range uses the proved native ordinary-zeta reflection consumer `classical_medium_typeI_witness_consumer_native`; it is not silently eliminated by the empty exceptional-zeta window. The Type-II consumer uses the actual fourth-moment reduction. The native source selects the maximum of the finitely many branch constants and a common maximum of their height thresholds, retaining the binwise analytic-multiplicity factor. This yields the positive-slab estimate. The half-line endpoint uses the separately proved global zero-count estimate.

Lean correspondence: `GuthMaynard.NativeMoment.ClassicalEndpointDensity.classical_endpoint_positive_slab_native,ingham_endpoint_positive_slab_native;MediumTypeIEndpoint;TypeIIFourthMomentReduction;ClassicalDensity.ingham_zero_density_at_half_native`.

Proposition 9.4 (The actual Ingham lower strip)**.** *For each $\sigma \in [1/2, 9/13]$ and $\varepsilon > 0$ there is $C > 0$ such that for all $T \geq 2$,*

$$N(\sigma, T) \leq C T^{3(1-\sigma)/(2-\sigma)+\varepsilon}.$$

Proof. For $\sigma > 1/2$ use the certificate just proved and Input 9.3, followed by the native dyadic-to-global transfer. The native bound is expressed as `EpsilonPowerBound`: for each positive ε the absolute count is Big-O at infinity of $T^{\varepsilon}|T^{p}|$. The native rectangle allows $\operatorname{Re}\rho \leq 1$ rather than $\operatorname{Re}\rho < 1$. Since $\sigma > 0$, every native zero has positive real part, and imported zeta nonvanishing for $\operatorname{Re}\rho \geq 1$ makes its upper inequality strict. Thus the finite native zero set equals $Z(\sigma; -T, T)$ exactly, and both sums use the same natural analytic order. At $\sigma = 1/2$ use the separate native global estimate, where $p = 1$. For either case a positive Big-O constant is obtained; because $p = 3(1-\sigma)/(2-\sigma) \geq 0$, the finite bounded-height range is absorbed by $Q = N(\sigma, R)$ with $R \geq 2$ above the eventual threshold. $C + Q$ works since $T^{p+\varepsilon} \geq 1$ for $T \geq 2$. This proves the all-height actual-count statement without changing multiplicity conventions. □

Lean correspondence: `GuthMaynard.Improvement.NativeIngham.ingham_lower_density;NativeDensitySourceBridge.native_N_eq_zetaDensityCount,native_density_epsilon_bound;NativeDensityConventionBridge`.

9.2 The physical twelfth moment and the high-value producer

Put $z(t) = |\zeta(1/2 + it)|$ and $E(H, V) = \{t \in [H, 2H] : z(t) \geq V\}$.

Established analytic input 9.5 (Native high-value finite-peak theorem)**.** For every $\eta > 0$, eventually in H, uniformly for $V > 0$, $V \geq H^{1/8+\eta}$, every finite one-separated $W \subset E(H, V)$ obeys

$$|W| V^{12} \leq H^{2+\eta}.$$

This actual-zeta input is supplied by the native point-mean producer and the native Atkinson stationary-source local-mean producer: `PointMean.pointMeanInput_native` and `Atkinson.localMeanStationaryInput_native`. Their packet width ranges, local means, and stationary transform estimates are substantive analytic foundations. They are not deduced in this document from a pointwise growth bound or an unnamed moment assumption.

Lean correspondence: `MathCollab.Density.Stronger.NativePeaks.pointValue_peak_card_native;PointMean.PeaksInput;Atkinson.AtkinsonNativeLocalMean;Peaks.WidthRanges`.

Exact point-mean and packet interfaces. Write $z(t) = |\zeta(1/2+\mathrm{i}t)|$. The proved point-mean interface selects $P > 0$ such that for $T \geq 10$, $V > 0$, $G, L \geq 0$, a finite one-separated $W \subset [c - G/2, c + G/2]$, with $10 \leq c - G/2$, $c + G/2 \leq T$, $(\log t)^2 \leq L$ on W, $G/2 + L \leq G$, and $z(t) \geq V$, one has

$$V^2|W| \leq P \log T \left(|W| + 4 \int_{c-G}^{c+G} z(u)^2 du \right).$$

Define the actual excess $E_G(t; e) = \max(0, \int_{t-G}^{t+G} z(u)^2 du - e)$. The proved packet interface, for every $\delta, \kappa, \nu > 0$, selects $C, D > 0$, $H_0 \geq 40000$ before all subsequent variables. If $H \geq H_0$, W is G-separated, $H \leq t \leq 2H$ and $t^\delta \leq G \leq t^{1/2-\delta}$, $t^{1/4+\kappa} \leq G$ on W, and $W \subset [A, A + L]$, it states

$$\left(\sum_{t \in W} E_G(t; CG \log t) \right)^2 \leq DH^\nu \left(|W|H/G + |W|^2 \sqrt{GL} \right).$$

These are the precise two imported analytic inequalities in the peak argument. Their producers use the actual Gamma–zeta-square point contour and the stationary Atkinson local mean, respectively.

Lean correspondence: `MathCollab.Density.Stronger.Peaks.Inputs:PointMeanInput,LocalMeanPacketInput;PointMean.Equation44.heathBrown_equation44_native`.

Finite packet absorption and peak aggregation. Given a packet coefficient $A_0 > 0$ and excess level $Y_0 > 0$, choose $L_0 = (Y_0^2/(4A_0))^2/G$. Then $2A_0\sqrt{GL_0} = Y_0^2/2 \leq Y_0^2$. For a localized set of size r, $rY_0 \leq \sum E_G$ and the packet inequality imply $r^2 Y_0^2 \leq A_0(rH/G + r^2\sqrt{GL_0})$. If $r > 0$, absorption and cancellation give $r \leq 2A_0H/(GY_0^2)$; if $r = 0$ the same bound is immediate. Partition $[H, 2H]$ into at most $\lfloor H/L_0 \rfloor + 1$ fibers. This gives the exact global bound

$$r \leq 2A_0H/(GY_0^2) + 32A_0^3H^2/Y_0^6.$$

Applying the packet interface with loss $\nu/3$ puts both terms below $D'H^\nu(H/(GY_0^2) + H^2/Y_0^6)$.

For point peaks, use half-width bins $n = \lfloor (t - H)/(G/2) \rfloor_+$ with center $c_n = H + nG/2$ and occupancy r_n. The point-mean input gives, with $\mathcal{L} = \log(3H)$, $V^2 r_n \leq P\mathcal{L}(r_n + 4I_n)$, where $I_n = \int_{c_n-G}^{c_n+G} z^2$. When $V^2 \geq 2P\mathcal{L}$, put $A_1 = V^2/(16P\mathcal{L})$; then $I_n \geq 2r_nA_1$. If the packet error is at most A_1, every bin with $r_n \geq m \geq 1$ has excess at least mA_1. Even and odd bin centers separately are G-separated. The preceding excess-count bound yields

$$|\{n : r_n \geq m\}| \leq 2D'H^\nu \left(\frac{H}{G(mA_1)^2} + \frac{H^2}{(mA_1)^6} \right) \leq \frac{2D'H^\nu}{m^2} \left(\frac{H}{GA_1^2} + \frac{H^2}{A_1^6} \right).$$

The finite occupancy identity $|W| = \sum_{m \geq 1} |\{n : r_n \geq m\}|$ and $\sum_{m \geq 1} m^{-2} \leq 2$ give the point-count bound at width G. Choose $K = 16PC + 1$ and $G = V^2/(K\mathcal{L}^2)$. The error

condition follows from $K \geq 16PC$; the point-mean width condition follows eventually from $2\mathcal{L}^2 \leq G$. Subject to

$$K\mathcal{L}^2(2H)^{1/4+\kappa} \leq V^2 \leq K\mathcal{L}^2 H^{1/2-\delta},$$

the exact width conditions hold throughout the slab, and substitution yields

$$|W| \leq D'' H^{\nu} \left(H\mathcal{L}^4/V^6 + H^2\mathcal{L}^6/V^{12} \right).$$

For the requested peak loss ε, the source chooses $\eta = \min(\varepsilon/16, 1/1000)$, $\delta = 1/48$, $\kappa = \nu = \eta$. For nonempty W, its actual Weyl-growth bound gives $V \leq H^{1/6+\eta}$. The lower threshold $V \geq H^{1/8+\varepsilon}$ and this growth bound make both width comparisons valid eventually. Multiplying by V^{12} gives

$$D''\mathcal{L}^4 H^{2+7\eta} + D''\mathcal{L}^6 H^{2+\eta} \leq H^{2+8\eta} \leq H^{2+\varepsilon}$$

eventually. Empty sets are handled directly. This proves the finite peak consumer used in Input 9.5, including every occupancy and width loss, relative to its two precisely stated analytic producers.

Lean correspondence: `MathCollab.Density.Stronger.AtkinsonCardinalityAbsorption;AtkinsonGlobalCardinality;Peaks.Clusters;Peaks.ClusterCount;Peaks.WidthRanges`.

Lemma 9.6 (High-value measure and the twelfth moment)**.** *For each $\varepsilon > 0$ the source proves*

$$\int_H^{2H} z(t)^{12} dt \ll_\varepsilon H^{2+\varepsilon} \quad \textit{eventually}, \qquad \int_0^{3T} z(t)^{12} dt \ll_\varepsilon T^{2+\varepsilon} \quad \textit{eventually}.$$

Proof. The finite separated-cover lemma applied to Input 9.5 gives $|E(H,V)| \leq 2H^{2+\eta}/V^{12}$ for $V \geq H^{1/8+\eta}$. For a requested exponent ε take $\eta = \varepsilon/20$. Below $V = H^{1/8+\eta}$, $z^{12} \leq V^8 z^4$, so Input 7.4 contributes at most $CH^{2+9\eta}$. For the high values the source uses its established actual growth bound $z^{12} \leq H^3$ on the dyadic interval eventually. With $\eta' < \eta$, the measure-tail bound at η' and layer integration of the twelfth power give

$$\int_{E(H,H^{1/8+\eta'})} z^{12} \leq 2H^{2+\eta'} \left(1 + \log \frac{H^3}{(H^{1/8+\eta'})^{12}} \right) \leq 2H^{2+\eta'}(1 + 3\log H).$$

The logarithm is absorbed into the positive exponent gap to $2+\eta$. The set with the larger cutoff $H^{1/8+\eta}$ is a subset, so its high contribution is at most $H^{2+\eta}$ eventually. Since $9\eta \leq \varepsilon$, summing the two parts proves the dyadic bound. The physical interval follows from the proved dyadic integral accumulation for the continuous nonnegative function z^{12}, with exponent $2+\varepsilon > 0$, absorbing the initial compact interval. Replacing the terminal height by $3T$ changes only the constant by $3^{2+\varepsilon}$. □

Lean correspondence: `MathCollab.Density.Stronger.ConditionalMoment.zeta_twelfth_dyadic_of_fourth_and_tail,zeta_twelfth_physical_of_dyadic;MomentGrowth;NativeMoment.zeta_twelfth_physical_native`.

9.3 The smooth detector family above three quarters

For a fixed $3/4 < \sigma < 1$ put

$$\alpha = 1/(2\sigma), \quad c = 3/(4\sigma) = 3\alpha/2, \quad q(\sigma) = 3(1-\sigma)/(2\sigma), \quad X = T^{\delta}, \quad Y = T^{c}.$$

We have $0 < c < 1$, and choose $\delta > 0$ before heights and zeros. The source's actual smooth dyadic partition is constructed from mathlib's smooth transition s, equal to zero on $(-\infty, 0]$ and one on $[1, \infty)$:

$$d(x) = 1 - s(4x - 5), \qquad \omega(x) = d(x) - d(2x).$$

It is nonnegative, at most one, and supported in $[5/8, 3/2] \subset (1/2, 2)$. For $x > 0$ the sum $\sum_{j\in\mathbb{Z}} \omega(x/2^j) = 1$ follows by telescoping: nonnegative scales sum to $1 - d(2x)$ and negative scales sum to $d(2x)$. The sum is pointwise finite by the support bounds. Splitting a block at its half-scale endpoints therefore introduces no endpoint contribution.

Lean correspondence: `MathCollab.Density.Stronger.SmoothPartition:dyadicStep,smoothDyadicWeight, hasSum_smoothDyadicWeight`.

Its actual block is

$$B_j(\rho) = \sum_{n\geq 2} c_X(n) n^{-\rho} \mathrm{e}^{-n/Y} \omega(n/2^j).$$

The finite scale set is $\mathcal{S}_R = \{j \in \mathbb{Z} : j \geq 0, 2^j \leq R\}$, $R = Y(\log T)^2$. A long block has $2^j > T^{c/2}$ and $|B_j(\rho)| \geq 1/(24 \log T)$. For the short family split a smooth block into natural dyadic half scales $Q_{j,e} = 2^{j_{\mathrm{nat}}-1+e}$ for $e = 0, 1$. The coefficient on a half is $c_X(n)\mathrm{e}^{-n/Y}\omega(n/2^j)$, retaining the original smooth scale 2^j. Apply the same real-part Taylor weights of Chapter 6 to each half. The Taylor index set is $\mathcal{S}_R \times \{0, 1\} \times \{0, \ldots, \lceil 10 \log T\rceil_+\}$. Let $K = \lceil 2\alpha/\delta\rceil_+$. Take convolution powers $1 \leq k \leq K$ and pieces $0 \leq \ell < k$, retaining only members with $T^\alpha \leq Q^k \leq T^c$. Their integer starts are $L_q = 2^\ell Q^k$, their coefficients are the conjugated convolution coefficients divided by T^η. This exactly describes `smoothPoweredIndices` and `smoothPoweredFamilyCoefficient`.

Lean correspondence: `MathCollab.Density.Stronger.SmoothDetector;ShortFixedFamily;ShortTaylorCover;ShortPoweredFamily`.

Established analytic input 9.7 (Native smooth detector cover and long-mean reduction)**.** For fixed $3/4 < \sigma < 1$, $0 < \delta \leq 1/8$, $\eta > 0$, eventually every actual zero in the positive slab is either covered by the preceding normalized powered short family at its original ordinate, with value $\geq L_q^\sigma T^{-2\eta}$, or has a long smooth block. The complementary, multiplicity-weighted long count N_{long} satisfies

$$N_{\mathrm{long}} Y^{6\sigma-3} \leq C T^{6\delta} (\log T)^{13} \left(\int_0^{3T} z(t)^{12} dt + T^{-113} \right). \tag{9.1}$$

What is used in the long reduction. For the complementary zero the source proves the lower bound

$$Y^{(\sigma-1/2)/2} \leq 48C\sqrt{X} \log T \int_{\mathbb{R}} \frac{z(\mathrm{Im}\,\rho + t)}{(1 + |t|)^{128}}\, dt.$$

It extracts separated representatives with local analytic multiplicity, raises this bound to the twelfth power, and applies Hölder and the physical weighted-mean packing lemma. The near weighted integral is controlled by the physical twelfth moment; the far tail by the proved zeta growth and the integrable weight of order 128. This yields the exact exponents X^6, $Y^{6\sigma-3}$, $(\log T)^{13}$, and remainder T^{-113} in (9.1). Long and short counts form an exact partition of the original slab, weighted by $m(\rho)$.

Lean correspondence: `MathCollab.Density.Stronger.ShortPoweredFamily.smooth_long_or_short_powered_cover;LongMultiplicityReduction.smoothLongZeroCount_physical_moment_reduction;WeightedCriticalMean;LongMeanCounting`.

Proposition 9.8 (Short part of the stronger density branch)**.** *If $\eta > 0$ and $8\eta < \sigma - 3/4$, the actual multiplicity count covered by the smooth powered family satisfies $N_{\mathrm{short}} \ll_{\sigma,\delta,\eta} T^{q(\sigma)+12\eta}$ eventually.*

Proof. The source establishes for every eligible member $T^{\alpha} \le L_q \le 2^K T^c \le T$ eventually, bounded normalized coefficients, and at most $144K^2(\log T)^2$ members. Use the earlier native integer large-values bound (the high-threshold input described after (4.2)) at comparison height $H_a = 32T(\log T)^2$. It requires $H_a^{1/3} \le L_q \le H_a$ and $L_q^{3/4+\kappa} \le L_q^{\sigma} T^{-2\eta}$. Take $\kappa = (\sigma - 3/4)/2 > 0$; the margin $8\eta < \sigma - 3/4$ implies $3/4 + \kappa \le \sigma - 4\eta$. Since $L_q \ge T^{\alpha} \ge \sqrt{T}$, these height and threshold conditions hold eventually exactly as in the source.

With $V_q = L_q^{\sigma} T^{-2\eta}$ the two terms are $L_q^{2-2\sigma} T^{4\eta}$ and $L_q^{3-4\sigma}\sqrt{H_a}\, T^{8\eta}$. The positive length exponent uses the upper length bound and $2c(1-\sigma) = q(\sigma)$; the negative one uses the lower bound and $1/2 + (3-4\sigma)\alpha = q(\sigma)$. Each member's budget is at most $(2^K + 32)\log T\, T^{q(\sigma)+8\eta}$. The indexed finite cover and multiplicity extraction add a factor $\log(2T) H_a^{\eta}$ and the number of family members. Eventually $H_a^{\eta} \le T^{2\eta}$, and the resulting four logarithms are at most $T^{2\eta}$. This gives exactly the total loss 12η. It bounds only the explicitly covered subset; the other zeros remain in N_{long}. □

Lean correspondence: `MathCollab.Density.Stronger.ShortExponentAccounting.smooth_family_sum_eventually;ShortZeroCount.smoothShortZeroCount_bound`.

Proposition 9.9 (The strictly upper density branch)**.** *For every fixed $\sigma > 3/4$ and $\varepsilon > 0$ there is $C > 0$ such that for all $T \ge 2$,*

$$N(\sigma, T) \le C T^{3(1-\sigma)/(2\sigma)+\varepsilon}.$$

Proof. For $\sigma \ge 1$ the count is zero, so take $C = 1$. For $3/4 < \sigma < 1$, put $d = 1 - c > 0$. Choose

$$\delta = \min(1/16, d/24), \quad \eta = \min(\varepsilon/24, (\sigma - 3/4)/16), \quad \nu = d/16.$$

Then $0 < \delta \le 1/8$, $\delta < d/12$, $8\eta < \sigma - 3/4$, $12\eta \le \varepsilon$, and $0 < \nu \le d/8$. Lemma 9.6 gives $\int_0^{3T} z^{12} \le DT^{2+\nu}$ eventually. Insert this in (9.1); its extra remainder is also at most $T^{2+\nu}$ for $T \ge 1$. The identity

$$2 + (6 - 12\sigma)c/2 = q(\sigma) - (1 - c)$$

shows that the raw long exponent is $q - d + 6\delta + \nu$. Because $6\delta < d/2$ and $\nu \le d/8$, a positive margin remains to $q - d/4$; the thirteen logarithms are absorbed into that margin. Thus $N_{\mathrm{long}} \ll T^{q-d/4}$ eventually. Proposition 9.8 gives $N_{\mathrm{short}} \ll T^{q+12\eta}$. Their exact partition gives an actual positive-slab bound with exponent $q + 12\eta > 0$. Apply Lemma 8.4 and finally monotonicity in the exponent to replace 12η by ε for $T \ge 2$. The choice $d > 0$ and the high-threshold margin fail at $\sigma = 3/4$; no endpoint at that line is inferred. □

Lean correspondence: `MathCollab.Density.Stronger.FinalParameters.exists_stronger_loss_parameters;ConditionalDensity.stronger_density_bound_of_twelfth_moment;NativeDensity.stronger_density_bound_native`.

9.4 The separate coefficient-two branch used in the global proof

The source's global-uniform theorem uses an earlier coefficient-two result above 3/4, not the stronger branch just proved. To preserve that actual proof route we record its exact established interface.

Established analytic input 9.10 (Native Weyl-based density conclusion)**.** The source proves, for $\sigma > 3/4$ and every $\varepsilon > 0$,

$$N(\sigma, T) \ll_{\sigma,\varepsilon} T^{2(1-\sigma)+\varepsilon} \quad (T \geq 2).$$

It discharges the actual critical-line Weyl input $z(t) \leq C_w(1+|t|)^{1/6+\eta_0}$ with $0 \leq \eta_0 < 1/48$.

Use of the imported machinery. The source chooses the positive counting loss $\min((\sigma - 3/4)/16, \varepsilon/24)$ before heights, obtains an eventual actual positive-slab bound with exponent $2(1-\sigma)$ plus twelve such losses, and uses the actual-count dyadic summation. The Weyl growth is proved by the native van der Corput, second-derivative, and explicit Weyl chain. The contour errors, powered family, separated counts, and analytic multiplicity restoration in that branch are proved dependencies, not unproved density premises. If $\sigma \geq 1$ it again uses the empty count.

Lean correspondence: `MathCollab.Density.DensityConclusion.zeta_density_bound_of_weyl;DensityTheorem.zeta_density_bound;GuthMaynard.WeylZeta;WeylPort`.

Chapter 10

Endpoint assembly, uniform constants, and the boundary audit

10.1 The fixed-line piecewise theorem

Proof of Theorem 1.6. If $1/2 \le \sigma \le 9/13$, apply Proposition 9.4. If $9/13 < \sigma \le 5/7$, use Corollary 8.5 with g. If $5/7 < \sigma \le 3/4$, use its h bound. If $3/4 < \sigma \le 1$, use Proposition 9.9. The resulting exponent is exactly $f(\sigma) = A(\sigma)(1-\sigma)$. The constant in each branch may depend on the fixed σ and ε; this case distinction makes no additional uniformity claim. □

Lean correspondence: `GuthMaynard.Improvement.CombinedDensityCurve.combined_density_curve;GuthMaynardStatement.coefficient_times_gap,full_curve`.

The endpoint values are

$$f(1/2) = 1, \quad f(9/13) = 12/17, \quad f(5/7) = 9/14, \quad f(3/4) = 9/16, \quad f(1) = 0.$$

At $9/13$ the two rational formulas coincide; at $5/7$ the rational formula equals $(9/4)(1-\sigma)$. The value at $3/4$ belongs to the closed upper window, not the strictly upper branch.

Lean correspondence: `GuthMaynard.Improvement.CombinedDensityCurve.combined_density_curve_endpoints`.

10.2 The actual route to the fixed-line coefficient 39/17

Let $c_0 = 39/17$. The source compares

$$\frac{3(1-\sigma)}{2-\sigma} \le c_0(1-\sigma) \quad (1/2 \le \sigma \le 9/13),$$

$$\frac{18(1-\sigma)}{7\sigma+3} \le c_0(1-\sigma) \quad (9/13 \le \sigma \le 5/7).$$

The first uses $2-\sigma \ge 17/13$, and the second $7\sigma+3 \ge 102/13$. Both denominators are positive and $1-\sigma \ge 0$. Above $3/4$ Input 9.10 fits since $2 < c_0$ and $1-\sigma \ge 0$. The source closes $[5/7, 3/4]$ through its *separate* padded upper-window route. It uses the same fixed powered detector family at lower exponent $3/4$, upper exponent $9/8$, but uses the fixed second split $7/8$ rather than $s(\sigma)$. The needed comparisons are

$$\tfrac{9}{4}(1-\sigma), \quad 39/14 - 3\sigma, \quad 63/20 - 7\sigma/2, \quad 15/8 - 7\sigma/4 \ \le c_0(1-\sigma).$$

They are affine in σ and hold at both endpoints $5/7$ and $3/4$. Padded time $S = H$ below $H^{5/6}$ and $S = N^{6/5}$ between $H^{5/6}$ and $H^{7/8}$ yields the first three budgets; the mean-value branch above $H^{7/8}$ yields the fourth. The loss is 10δ as in Chapter 7. The fixed-family cover adds η, the unconditional weak Type-II exponent fits $c_0(1-\sigma)$, and Jensen restoration plus dyadic summation gives the actual upper density bound. Thus the whole fixed-line c_0 theorem has no `GuthMaynardUpperDensity` or `ShortMobiusFourthMoment` premise.

Lean correspondence: `GuthMaynard.Improvement.UpperWindowBudgets;UpperWindowLargeValues;UpperTypeICount;WeakUpperDensity.upper_density;GlobalDensityIngham.global_density_fixed_sigma_of_ingham`.

10.3 Finite-grid uniformization

Lemma 10.1 (One-sided-modulus uniformization)**.** *Let $a < b$, $K > 0$, and $q : [a, b] \to \mathbb{R}$ satisfy*

$$q(x) \le q(y) + K(y - x) \quad (a \le x \le y \le b).$$

Suppose each $\sigma \in [a, b]$ and each $\eta > 0$ has a constant with $N(\sigma, T) \le C_{\sigma,\eta} T^{q(\sigma)+\eta}$ for all $T \ge 2$. For every $\varepsilon > 0$ a single positive C then works for all σ, T: $N(\sigma, T) \le C T^{q(\sigma)+\varepsilon}$.

Proof. Choose a natural $n > 2K(b-a)/\varepsilon$. Set $x_j = a + j(b-a)/n$ for $j = 0, \dots, n$. At each grid point choose $C_j > 0$ with the fixed-line estimate at loss $\varepsilon/2$. Let $C = 1 + \sum_{j=0}^{n} C_j$. For a given σ, let $j = \lfloor (\sigma - a)n/(b-a) \rfloor_+$; then $0 \le j \le n$, $x_j \le \sigma$ and $\sigma - x_j \le (b-a)/n$. The one-sided modulus gives $q(x_j) + \varepsilon/2 \le q(\sigma) + \varepsilon$. Monotonicity of the actual zero count in its real-part cutoff and monotonicity of T-powers in their exponent, for $T \ge 2$, give

$$N(\sigma, T) \le N(x_j, T) \le C_j T^{q(x_j)+\varepsilon/2} \le C_j T^{q(\sigma)+\varepsilon} \le C T^{q(\sigma)+\varepsilon}.$$

The constant was selected from a finite grid before σ, T. □

Lean correspondence: `GuthMaynard.Improvement.CurveUniformization.density_curve_uniform_on_interval`.

Proof of Theorem 1.7. Use the actual fixed-line coefficient-c_0 proof from the preceding section. The source specializes finite-grid uniformization to $q(\sigma) = c_0(1 - \sigma)$ on $[1/2, 1]$. More explicitly choose $n > c_0/\varepsilon$ and $x_j = 1/2 + j/(2n)$, using fixed-line loss $\varepsilon/2$. Then $c_0(\sigma - x_j) \le c_0/(2n) \le \varepsilon/2$. The constant $1 + \sum C_j$ and the exact comparison in Lemma 10.1 give the assertion, including $\sigma = 1$. This proof does not attempt to uniformize the discontinuous exact curve. □

Lean correspondence: `GuthMaynard.Improvement.ShortDensityUniform.zetaDensityCount_uniform_of_fixed_sigma;GlobalDensityIngham.global_density_uniform_of_ingham;GlobalDensity.global_density_uniform;GuthMaynardStatement.global_uniform`.

10.4 Uniformity on the exact left curve and away from the strict join

The source proves each smooth branch has a one-sided modulus with $K = 4$. For $1/2 \le x \le y \le 9/13$,

$$\frac{3(1-x)}{2-x} - \frac{3(1-y)}{2-y} = \frac{3(y-x)}{(2-x)(2-y)} \le 4(y-x).$$

For $9/13 \le x \le y \le 5/7$,

$$\frac{18(1-x)}{7x+3} - \frac{18(1-y)}{7y+3} = \frac{180(y-x)}{(7x+3)(7y+3)} \le 4(y-x),$$

using $(7x+3)(7y+3) \ge 49$. For the linear branch the difference is $(9/4)(y-x) \le 4(y-x)$. For $3/4 \le x \le y \le 1$ the upper rational expression has

$$\frac{3(1-x)}{2x} - \frac{3(1-y)}{2y} = \frac{3(y-x)}{2xy} \le 4(y-x), \qquad xy \ge 9/16.$$

Only this modulus statement includes $x = 3/4$; the stronger density theorem itself still requires strict inequality.

Lean correspondence: `GuthMaynard.Improvement.CurveModulus`.

Proposition 10.2 (Uniform exact left curve)**.** *For every $\varepsilon > 0$ one constant works for all $\sigma \in [1/2, 3/4]$ and $T \ge 2$ in the bound $N(\sigma, T) \le CT^{f(\sigma)+\varepsilon}$.*

Proof. Apply Lemma 10.1 separately to the three closed intervals $[1/2, 9/13]$, $[9/13, 5/7]$, and $[5/7, 3/4]$, using the branch fixed-line bounds and the preceding moduli. The sum of the three positive constants works everywhere on their union. The definitions agree at the two interior joins, so there is no endpoint discrepancy. □

Lean correspondence: `GuthMaynard.Improvement.FullCurveLeftUniform.full_density_curve_uniform_left`.

Proof of Theorem 1.8. Let C_1 be the left constant from Proposition 10.2. If $3/4 + \delta < 1$, apply Lemma 10.1 to $[3/4 + \delta, 1]$ with $q(\sigma) = 3(1-\sigma)/(2\sigma)$ and $K = 4$. Proposition 9.9 applies at every point of that interval, so a right constant C_2 exists. Take $C_1 + C_2$ and choose the appropriate branch. If $3/4 + \delta \ge 1$, every admitted point in the right alternative has $\sigma \ge 1$, so its count is zero. The left constant alone works. This handles intervals that degenerate to the single endpoint or are empty. □

Lean correspondence: `GuthMaynard.Improvement.CombinedDensityCurve.combined_density_curve_uniform_away_boundary;GuthMaynardStatement.uniform_away_boundary`.

10.5 A boundary obstruction is not a counterexample to uniform density

Proposition 10.3 (Exact boundary-modulus obstruction)**.** *For every real K there is $\sigma \in (3/4, 1]$ such that*

$$f(\sigma) + K(\sigma - 3/4) < f(3/4).$$

Proof. For every $\sigma > 3/4$, the right formula satisfies $f(\sigma) = 3(1-\sigma)/(2\sigma) < 1/2$, by multiplying by $2\sigma > 0$. Set $\delta_K = 1/(64(|K|+1))$ and $\sigma = 3/4 + \delta_K$. Then $0 < \delta_K \le 1/64$, $\sigma \le 1$, and $K\delta_K \le 1/64$. Thus $f(\sigma) + K\delta_K < 1/2 + 1/64 = 33/64 < 9/16 = f(3/4)$. □

Lean correspondence: `GuthMaynard.Improvement.CurveBoundaryAudit.combined_curve_right_lt_half,combined_curve_boundary_modulus_obstruction`.

It follows that the one-sided finite-K modulus required by Lemma 10.1 cannot hold across $3/4$ for the combined curve. This proves a limitation of that uniformization method, not the falsehood of a whole-strip exact-curve uniform density estimate.

10.6 The independent actual-zero assertion in the interface

The endpoint specification states actualZeroBound(σ, T, B) as existence of a finite set Z whose membership is exactly

$$\zeta(\rho) = 0, \quad 0 < \operatorname{Re}\rho < 1, \quad \sigma \le \operatorname{Re}\rho, \quad |\operatorname{Im}\rho| \le T,$$

and for which $\sum_{\rho \in Z} m(\rho) \le B$. Given any bound on $N(\sigma, T)$, choose $Z = Z(\sigma; -T, T)$. The finite-set theorem from Chapter 1 proves its existence, and $-T \le \operatorname{Im}\rho \le T$ is equivalent to $|\operatorname{Im}\rho| \le T$. The sum defining N is exactly the required sum, so each endpoint count bound yields the independent assertion. The coefficient identity $A(\sigma)(1-\sigma) = f(\sigma)$ is verified branch by branch. The large-values specification is definitionally the same bounded polynomial and the same middle-range proposition proved in Chapter 5. These are all the final wrapper steps.

Lean correspondence: `GuthMaynardStatement.actualZeroBound_of_count`, `coefficient_times_gap`, `middle_large_valuesinlean/Solution.lean`.

Appendix A

Dependency ledger, provenance, and verification boundary

A.1 The mathematical inputs used, not additional endpoint assumptions

The inputs below are established theorems in the pinned source closure. Their use is explicit in the preceding proofs. A future paper expanding all of their internal analytic proofs would require additional chapters; this blueprint does not claim such an expansion has already been made.

1. **Analytic zero foundations.** mathlib analytic zeta away from 1, nonvanishing in $\operatorname{Re} s \geq 1$, isolated zeros on compact sets, the identity theorem, finite analytic order and local factorization. Used to make the zero count nonvacuous and multiplicities genuine.

2. **Ordinary Fourier foundations.** Lebesgue/Bochner integration, Tonelli/Fubini under the stated integrability, normalized Plancherel, Fourier inversion for the cut signals, convolution regularity, rapid decay of compact smooth/Schwartz inputs, Poisson summation with explicit scale. Used by affine energies, source sampling, the masked cubic, and extraction.

3. **Affine native descent.** Source Lemma-9.2 recurrence, divisor ray counting and retained Poisson budgets, adaptive smoothing, and finite natural-number descent. The height profile, support-depth budget, and unequal RK factor are retained in Chapter 2.

4. **Nuclear operator foundations.** Positive rank-one sums and integrals, finite-dimensional eigenbases, witness/Bessel bounds, Schur operator norm, cubic and mixed trace limits through finite compressions. No infinite eigenbasis or commuting correction operator is assumed.

5. **Native classical large values.** The high-threshold integer estimate is applied only with its $\kappa > 0$ and scale restrictions; the separated mean square closes low thresholds and real-length rounding.

6. **Actual contour foundations.** Möbius cancellation, Mellin inversion, regularized zeta poles, rectangle residues, Gamma strip decay, and uniform ordinate-tail packing. Used with independent X, Y.

7. **Native fourth moment.** `Fourth.zeta_fourth_dyadic` feeds the weak mollified fourth moment and the high/low twelfth-moment assembly. The sharper `ShortMobiusFourthMoment` contract remains separate and is not used in the unconditional endpoints.

8. **Native high-value peaks.** Point-mean and stationary-source Atkinson producers plus packet width ranges feed the actual finite peak bound. That bound feeds the high twelfth-moment tail, not an assumed unrestricted twelfth-moment conjecture.

9. **Native Ingham endpoint.** Classical detector dichotomy and the separate medium ordinary-zeta Type-I reflection consumer, coefficient normalization, bounded shifts with reseparation, Type-II fourth moment, and exact native-count bridge. The half endpoint has its own proof.

10. **Earlier coefficient-two upper density.** Explicit native Weyl input, detector error estimates, finite powered families, multiplicity restoration, and dyadic accumulation. It is the actual upper branch used by `GlobalDensityIngham`, even though the combined fixed-line curve has a stronger strictly upper branch.

A.2 Proof route and compatibility names

The final registered path is

> `GuthMaynardStatement.full_curve` follows from `combined_density_curve`.
> `GuthMaynardStatement.global_uniform` follows from `global_density_uniform`.
> `GuthMaynardStatement.uniform_away_boundary` follows from `combined_density_curve_uniform_away_boundary`.
> `GuthMaynardStatement.middle_large_values` follows from `middle_range_large_values`.

The wrappers are in the file `Solution.lean`; its endpoint namespace is `GuthMaynardStatement`. Similarly "candidate", "conditional", and "gate" in an internal filename are not evidence of an unproved endpoint. Each explicit intermediate contract must be checked where its producer is supplied. The internal `GuthMaynardUpperDensity` proposition is not a premise of `global_density_uniform`; `InghamLowerDensity` is discharged by `ingham_lower_density`; the final short branch uses `WeakShortDensity.short_density`.

A.3 Preserved attribution

New material in the repository carries AGPL-3.0-only licensing; imported ports retain their original notices. The pinned record identifies adapted material from Scott McColm at revisions `6e2d10c6c2252ee1575bb7ef12dea12e2f1a3af1` (MIT-0) and `2ace9e7c09a69fdcd1edae1ab6deb7cb3b4df1be` (MIT), selected Fourier, Poisson, Mellin and Gamma developments from Conor Grogan's `prime-minor-arcs-2-15` at `f369f267b4dcfebf010c8e7baa2c9602e2960eba` (Apache-2.0), and analytic foundations from `PrimeNumberTheoremAnd` at `4ecb950126c4290293c5662dfe0e884123171df5` (Apache-2.0). The original repository's `THIRD_PARTY_NOTICES.md`, `licenses/`, source headers, and `third_party/` manifests are preserved in the archive. The informal reconstruction is not attributed to those authors as an approved text.

A.4 What has and has not been checked here

The downloaded statement and solution files match the registry's SHA-256 values respectively:

> `b0feb005a241a3d2da02def8c8ae83cb9db2e743b6c5679245a5c22775e7675a`
> `8b3e9e2dea3baa6c9db288f6d6bc64662b8fd1f14525599e99abb7fd9bc9f7cb`.

The source was read and the displayed mathematics was reconstructed and cross-checked against the cited declarations. The LaTeX build and references are checked separately in the accompanying validation record. This does not constitute human peer review, an author endorsement, a novelty certificate, or a fresh complete kernel replay of the pinned Lean development. The registry's mechanical certificate and the repository's earlier local verification receipt have distinct provenance; neither is silently replaced by a claim that this reconstruction performed a full rebuild.

A.5 Source references

Bibliography

[1] Palomar Registry, *Zero-density estimates from an improved large-values bound*, entry PALOMAR-2026-10-11-000010, version 1. https://palomar-registry.org/entry?id=PALOMAR-2026-10-11-000010&version=1.

[2] M. Lewko, *zeta-density-39-over-17-lean*, fixed repository snapshot. https://github.com/mrklwk/zeta-density-39-over-17-lean/tree/ee2a8a32be49a4f73c7386e9dffdea24d7e3d8c0.

[3] The mathlib community, *mathlib4*, revision 065356127b1dc0016f66b7283ce0ce2c4055aa55. https://github.com/leanprover-community/mathlib4/tree/065356127b1dc0016f66b7283ce0ce2c4055aa55.

[4] The pinned source, *THIRD_PARTY_NOTICES.md* and source-identity manifests, with the licenses retained for each port.